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Inventor(s)

Radhakrishna; Ujwal et al.

SEMICONDUCTOR DEVICE WITH DRAIN ELECTRICAL CONTACT FORMING JUNCTIONS HAVING DIFFERENT ENERGY BARRIER HEIGHTS TO DRAIN LAYER

Abstract

A semiconductor device, such as a GaN-based high electron mobility transistor (HEMT), includes a hybrid drain contact structure over a channel layer and a barrier layer. The hybrid drain contact structure includes a first drain contact electrically coupled to the channel layer, a semiconductor layer over the barrier layer and including a first semiconductor portion and a second semiconductor portion, and a second drain contact on the semiconductor layer and electrically coupled to the first drain contact. The second drain contact includes a first metal portion and a second metal portion. The first metal portion and the first semiconductor portion form a first junction having a first energy barrier height. The second metal portion and the second semiconductor portion form a second junction having a second energy barrier height lower than the first energy barrier height.

Inventors: Radhakrishna; Ujwal (San Jose, CA), Tang; Zhikai (Sunnyvale, CA), Joh; Jungwoo (Allen, TX)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

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Background/Summary

BACKGROUND

[0001] A high electron mobility transistor (HEMT) may include a heterojunction formed using different semiconductor materials, where a channel may be formed near the heterojunction and between a source region and a drain region, and the channel may be turned on or off by applying an appropriate voltage level on a gate structure. Gallium nitride (GaN)-based HEMT devices generally have high breakdown field, high electron mobility, low resistance, high current, faster-switching speed, high thermal conductivity, and excellent reverse-recovery performance, and thus may be suitable for applications where a low-loss and high-efficiency performance is desired, such as power electronics, radio frequency (RF) circuits, etc.

SUMMARY

[0002] This Summary is provided to introduce examples of disclosed concepts in a simplified form, which are further described below in the Detailed Description including the drawings provided.

[0003] According to certain aspects, a semiconductor device may include a substrate having a channel layer and a barrier layer over the channel layer, a source contact electrically coupled to the channel layer, a gate structure over the barrier layer, and a drain contact structure. The drain contact structure includes a first drain contact electrically coupled to the channel layer, a semiconductor layer over the barrier layer and including a first semiconductor portion and a second semiconductor portion, and a second drain contact on the semiconductor layer and electrically coupled to the first drain contact. The second drain contact includes a first metal portion and a second metal portion, where the first metal portion and the first semiconductor portion form a first junction having a first energy barrier height, and the second metal portion and the second semiconductor portion form a second junction having a second energy barrier height lower than the first energy barrier height.

[0004] According to certain aspects, a method may include forming a patterned semiconductor layer over a barrier layer that is over a channel layer formed on a semiconductor substrate. The patterned semiconductor layer includes a plurality of regions, where each of the plurality of regions includes a first semiconductor portion and a second semiconductor portion. The method also includes forming first metal contacts on regions of the channel layer and first semiconductor portions of the plurality of regions of the patterned semiconductor layer, where the first semiconductor portions and the first metal contacts on the first semiconductor portions form first junctions. The method further includes depositing second metal contacts on second semiconductor portions of the plurality of regions of the patterned semiconductor layer, where the second semiconductor portions and the second metal contacts on the second semiconductor portions form second junctions having energy barrier heights greater than energy barrier heights of the first junctions.

[0005] The foregoing summary outlines rather broadly various features of examples of the present disclosure in order that the following detailed description may be better understood. Additional features and advantages of such examples will be described hereinafter. The described examples may be readily utilized as a basis for modifying or designing other examples that are within the scope of the appended claims.

[0006] This summary is neither intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire

specification of this disclosure, any or all drawings, and each claim. The foregoing, together with other features and examples, will be described in more detail below in the following specification, claims, and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Illustrative examples are described in detail below with reference to the following figures.

[0008] FIG. 1 is a cross-sectional view of an example of an enhancement mode high electron mobility transistor (HEMT) with an overlaid circuit model of the HEMT.

[0009] FIG. 2 is a cross-sectional view of another example of an enhancement mode HEMT with an overlaid circuit model of the HEMT.

[0010] FIG. 3A includes top-down views of an example of an HEMT including a hybrid drain contact structure.

[0011] FIG. 3B includes top-down views of another example of an HEMT including a hybrid drain contact structure.

[0012] FIG. 3C is a cross-sectional view of an example of a cross-section of the HEMT of FIG. 3A or 3B.

[0013] FIG. 3D is a cross-sectional view of another example of a cross-section of the HEMT of FIG. 3A or 3B.

[0014] FIG. 3E is a cross-sectional view of an example of a hybrid drain contact structure including a Schottky contact, with an overlaid circuit model of the hybrid drain contact structure.

[0015] FIG. 4 includes a graph illustrating reduction of the dynamic on-state resistance (dRon) of an HEMT using a hybrid drain contact.

[0016] FIG. 5A is a cross-sectional view of an example of an HEMT including a hybrid drain contact structure according to certain embodiments.

[0017] FIG. 5B illustrates examples of a hybrid drain contact structure.

[0018] FIG. 5C is a cross-sectional view of an example of a hybrid drain contact structure, with an overlaid circuit model of the hybrid drain contact structure.

[0019] FIG. 6A illustrates an example of an enhancement mode HEMT including a hybrid drain contact structure having a Schottky contact and an ohmic contact.

[0020] FIG. 6B illustrates an example of a depletion mode HEMT including a hybrid drain contact structure having a Schottky contact and an ohmic contact.

[0021] FIG. 7A illustrates an example of an enhancement mode HEMT including a hybrid drain contact structure having Schottky contacts of different barrier heights.

[0022] FIG. 7B illustrates an example of a depletion mode HEMT including a hybrid drain contact structure having Schottky contacts of different barrier heights.

[0023] FIG. 8A illustrates an example of an enhancement mode HEMT including a hybrid drain contact structure having Schottky contacts of different barrier heights and an ohmic contact.

[0024] FIG. 8B illustrates an example of a depletion mode HEMT including a hybrid drain contact structure having Schottky contacts of different barrier heights and an ohmic contact.

[0025] FIGS. 9A, 9B, and 9C illustrate an example of a drain region of an HEMT including a hybrid drain contact structure.

[0026] FIGS. 9D and 9E illustrate another example of a drain region of an HEMT including a hybrid drain contact structure.

[0027] FIGS. 10A, 10B, and 10C illustrate another example of a drain region of an HEMT including a hybrid drain contact structure.

[0028] FIGS. 10D and 10E illustrate yet another example of a drain region of an HEMT including a hybrid drain contact structure.

[0029] FIGS. 11A, 11B, 11C, 11D, 11E, 11F, 11G, and 11H illustrate examples of hybrid drain contact structures of HEMTs including a continuous semiconductor layer.

[0030] FIG. 12A illustrates an example of an enhancement mode HEMT including a hybrid drain contact structure.

[0031] FIG. 12B illustrates an example of an enhancement mode HEMT including a hybrid drain contact structure.

[0032] FIG. 12C illustrates another example of an enhancement mode HEMT including a hybrid drain contact structure.

[0033] FIGS. 13A, 13B, 13C, and 13D illustrate an example of a process of fabricating an HEMT including a hybrid drain contact structure.

[0034] FIG. 14 includes a flowchart illustrating an example of a process of fabricating an HEMT including a hybrid drain contact structure.

[0035] The drawings, and accompanying detailed description, are provided for understanding of features of various examples and do not limit the scope of the appended claims. The examples illustrated in the drawings and described in the accompanying detailed description may be readily utilized as a basis for modifying or designing other examples that are within the scope of the appended claims. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated may be employed without departing from the principles, or benefits touted, of this disclosure. Identical reference numerals may be used, where possible, to designate identical elements that are common among drawings. The figures are drawn to clearly illustrate the relevant elements or features and are not necessarily drawn to scale.

DETAILED DESCRIPTION

[0036] The present disclosure relates generally to a semiconductor device including a drain contact structure that includes junctions with different energy barrier heights. In some examples described herein, the semiconductor device includes a high electron mobility transistor (HEMT), such as a gallium nitride (GaN)-based enhancement mode or depletion mode HEMT. The junctions with the different energy barrier heights may be used to improve hole injection during switching at high drain voltage, mitigate current collapse, and reduce dynamic and static on-state resistance. Other benefits and advantages may also be achieved, such as simplifying layout, improving breakdown performance, and the like.

[0037] A GaN-based HEMT may include a heterojunction formed by a channel layer (e.g., a GaN layer) and a barrier layer (e.g., an aluminum gallium nitride (AlGaN) layer). A conductive channel can be formed by high-density two-dimensional electron gas (2DEG) at the heterojunction. For example, the 2DEG can have a sheet charge density greater than about 10^{13} cm^{-2} , and thus can have a very low on-state resistance. GaN-based HEMTs are attractive for high frequency and high power applications due to, for example, high breakdown field, high electron mobility, low static resistance, and high thermal conductivity of GaN-based HEMTs. However, GaN-based transistors may suffer from current collapse, which is an increase in the dynamic on-state drain-to-source resistance ($R_{\text{DS(on)}}$) during high-voltage switching operations. Current collapse may result in, for example, power loss, temperature increase, and reliability issues.

[0038] Several factors may contribute to the current collapse. For example, when a lateral HEMT device is in the off-state with a high drain bias, the large negative gate-to-drain bias and thus high electric field can lead to electron injection and trapping at regions such as the heterojunction interface and barrier layer surface, while the large positive drain-to-substrate voltage may cause electrons to be injected from the substrate and trapped in the buffer stack between the channel layer and the substrate. In addition, when the HEMT device operates in high voltages (e.g., a few hundred volts), a hard switching can generate hot electrons in the channel, which may be injected and trapped, for example, in the dielectric, at the surface, in the GaN channel, and/or in the buffer. The trapped negative charges may at least partially deplete carriers (e.g., electrons) in the channel,

causing a reduction in the channel carrier density and thus an increase in the channel resistance and a reduction in the current between the drain and the source ($I_{sub,DS}$). Electron trapping at the interface and border regions beneath the gate electrode may also cause a positive shift in the threshold voltage. When the HEMT device is switched on, it may take a considerable amount of time to release the electrons from surface states and buffer traps so that the drain-to-source resistance may decrease.

[0039] A hybrid drain contact may be used to mitigate the electron trapping and current collapse effects. For example, a hybrid drain contact structure may include, in addition to the drain contact structure that forms an ohmic contact with the channel layer, a p-type drain contact structure that includes a p-doped semiconductor layer (e.g., a p-doped GaN (p-GaN) layer) positioned below a drain contact (e.g., a metal layer). When the voltage level applied to the drain contact is high and thus the voltage drop in the channel layer is high, a diode formed by the p-doped semiconductor layer and the underlying channel layer may be turned on to inject holes into the barrier layer, the channel layer, and/or the buffer layer, to neutralize the trapped electrons at the off-state and during the switching. In order to avoid depleting the 2DEG channel in the drain region by the p-doped semiconductor layer, the barrier layer (e.g., an AlGaIn layer) under the p-doped semiconductor layer of the p-type drain contact structure may be thicker than the barrier layer under the p-doped semiconductor layer at the gate of an example of an enhancement mode HEMT (e-HEMT) that may be off without a positive gate voltage, and/or the p-doped semiconductor layer of the p-type drain contact structure may be thinner than the p-doped semiconductor layer at the gate of the enhancement mode HEMT. Therefore, fabricating normally-off GaN-based transistors (e.g., enhancement mode HEMTs, or e-HEMTs) with p-type drain contact structures may include growing the barrier layer or the p-GaN layer twice (or selective etching) to attain different barrier layer and/or p-doped semiconductor layer thicknesses at the gate and the drain. The additional processing steps may add significant costs and reduce yield/throughput.

[0040] In some enhancement mode GaN-based HEMTs, multiple p-type drain contact structures may be used to inject holes to neutralize the trapped electrons, thereby mitigating the current collapses. The barrier layer under the p-GaN layer of the p-type drain contact structures may have the same thickness as the barrier layer under the p-GaN layer at the gate of the enhancement mode HEMT, such that one epitaxial growth process may be used to grow the barrier layer to reduce the manufacturing cost. The p-GaN layer of the p-type drain contact structures and the p-GaN layer of the gate structure may also be fabricated using the same processes and may have the same thickness. Each of the multiple p-type drain contact structures may be physically separated from the remaining p-type drain contact structures, such that the 2DEG channel may be depleted mostly/only at certain regions at the drain, whereas other regions of the 2DEG channel at the drain are not depleted and thus may allow current to pass through. In this way, the current collapse may be reduced due to the hole injection by the multiple p-type drain contact structures, and the fabrication cost may also be reduced.

[0041] However, using the multiple separate p-type drain contact structures in GaN-based HEMTs may effectively reduce the channel width and increase the static $R_{sub,DS}$. In addition, the p-type drain contact structures may include Schottky junctions formed by the metal contact and the underlying p-GaN layer. A Schottky junction may reduce the leakage current during operations of the e-HEMT because the Schottky junction may be reverse biased when a positive voltage is applied to the gate or drain metal contact. However, the Schottky junction may result in a back-to-back diode configuration, which in turn may result in a floating, quasi-neutral node in the p-GaN layer. The back-to-back diodes may form a voltage divider, which may reduce the effective drive voltage, increase drain-to-source on-state resistance, and reduce drain current (I_d). The reduction in the effective drive voltage at the drain may result in a decrease in the hole injection capability of the p-type drain contact structure, and thus the p-type drain contact structure may be less effective in mitigating the current collapse. In switching events, the floating, quasi-neutral node may be

biased based on past switching events due to its floating nature, and thus may have a memory effect. This biasing that depends on past switching events may result in non-deterministic behavior of the HEMT and may adversely affect the drain current of the HEMT. In addition, due to the reduced hole injection capability, the dynamic on-state resistance ($dR_{sub.on}$) of the HEMT during the switching (e.g., immediately after turning on the HEMT) may be higher, and the switching time may be longer before the on-state resistance of the HEMT settles to a static on-state resistance.

[0042] Examples of a hybrid drain contact structure that can address at least some of the issues above are disclosed herein. According to some examples, a hybrid drain contact structure includes both an ohmic contact coupled to the channel layer and a p-type drain contact structure. The p-type drain contact structure includes two or more junctions with different energy barrier heights formed by one or more metal materials and a p-type semiconductor layer (e.g., a p-GaN layer) over the barrier layer. In one example, the p-type drain contact structure includes a first metal portion and a second metal portion. The first metal portion forms a first junction with a first portion of the p-type semiconductor layer, and the second metal portion forms a second junction with a second portion of the p-type semiconductor layer. The energy barrier height of the first junction is different from the energy barrier height of the second junction. For example, the first junction may be a high-barrier Schottky (HBS) junction and the second junction may be a low-barrier junction such as a low-barrier Schottky (LBS) junction having a lower barrier height than the HBS junction, or an ohmic junction. In some examples, a LBS junction has a barrier height lower than 1.7 eV, and an HBS junction has a barrier height higher than 1.7 eV.

[0043] The ohmic junction and/or low-barrier junction may provide a Kelvin connection to the p-type semiconductor layer to remove the floating, quasi-neutral node and reduce the voltage drop between the drain electrical contact and the floating, quasi-neutral node. Reducing the voltage drop may improve the hole injection efficiency of the p-type drain contact structure for mitigating current collapse and reducing dynamic on-state resistance dR_{on} . Due to the improved hole injection efficiency of the p-type drain contact structure, the p-type drain contact structure can be smaller to reduce the depleted region of the channel layer under the p-type drain contact structure (and the reduced channel width penalty), thereby reducing the static on-state resistance RD_{SON} . As such, the channel length would not need to be reduced in order to reduce the static on-state resistance RD_{SON} , and thus the breakdown voltage of the device may not be reduced. The smaller p-type drain contact structures may also simplify the layout. In addition, the voltage drop at the drain can be the same under DC conditions and in switching events. Therefore, the dynamic on-state resistance may be closer to the static on-state resistance.

[0044] In some examples of the p-type drain contact structure disclosed herein, the two or more junctions having different energy barrier heights may be implemented using a p-type semiconductor layer having a uniform composition and two or more different metal materials with different metal work functions. In some examples of the p-type drain contact structure, the two or more junctions having different energy barrier heights may be implemented using a same metal material and a same base semiconductor material (e.g., p-GaN) with different constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation in two or more different regions.

[0045] The hybrid drain contact structures disclosed herein may be used in enhancement mode HEMTs and depletion mode HEMTs. In some examples, an HEMT may include multiple p-type drain contact structures formed on separate regions of the p-type semiconductor layer. In some examples, the second metal portions of the multiple p-type drain contact structures and the drain contact metal that form the ohmic contact with the channel layer may include the same metal material, and may be parts of a continuous metal layer. In some examples, an HEMT disclosed herein may include one p-type drain contact structure that includes a continuous p-type semiconductor region, a continuous first metal portion, and a continuous second metal portion, where the barrier layer under the gate structure may have a lower thickness than the barrier layer

under the p-type drain contact structure, and/or the p-type semiconductor layer in the gate structure may have a higher thickness than the p-type semiconductor layer in the p-type drain contact structure, in order to achieve a low on-state resistance. In such examples, the layout may be less complex, but additional fabrication processes (e.g., epitaxial growth or etching) may be needed. [0046] Various features are described hereinafter with reference to the figures. An illustrated example may not have all the aspects or advantages shown. An aspect or an advantage described in conjunction with a particular example is not necessarily limited to that example and can be practiced in any other examples even if not so illustrated or if not so explicitly described. Further, methods described herein may be described in a particular order of operations, but other methods according to other examples may be implemented in various other orders (e.g., including different serial or parallel performance of various operations) with more or fewer operations.

[0047] Various examples are described herein. Although the specific examples may illustrate various aspects of the above generally described features, examples may incorporate any combination of the above generally described features (which are described in more detail in examples below). Three dimensional x-y-z axes are illustrated in some figures for ease of reference. Some cross-sectional views of various semiconductor devices herein may be general depictions to communicate various aspects or concepts concerning such semiconductor devices. More specifically, some drain contact structures illustrated in cross-sectional views may not necessarily accurately depict a structure of such drain contact contacts, except to the extent described herein. The illustrations of those drain contact structures are to communicate various aspects or concepts concerning those drain contact structures.

[0048] Various examples are described in the context of an HEMT. Some examples may be implemented in enhancement mode lateral HEMTs that are for high voltage (e.g., about 650 V to about 1,200 V) applications or low to medium voltage (e.g., about 10 V to about 100 V, or about 10 V to about 200 V) applications. In other examples, the semiconductor device may include a bidirectional field effect transistor (FET), a gated Schottky barrier diode (e.g., gate-to-drain shorted structure or gate-to-source shorted structure), or similar devices. Some examples may be implemented with any epitaxial structure, any field plate and/or ohmic contact structure, a planar or three-dimensional structure (e.g., fin structure), and/or various other modifications.

[0049] For the sake of illustration, some of the examples disclosed herein may focus on group-III nitride-based devices, such as GaN-based HEMTs. However, this disclosure is not limited to GaN-based HEMTs and can be applied to other devices that include heterostructures formed by other semiconductor materials, such as other group-III nitride or other III-V semiconductor materials, where the heterostructures may induce 2DEG at the heterojunction interface.

[0050] GaN-based HEMTs include heterostructures that induce two-dimensional electron gas (2DEG) at the interface between two GaN-based materials with different bandgaps. In one example, the heterostructure may be formed by a GaN layer and an $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.(1-x)}}\text{N}$ layer, where x is the concentration of aluminum. The GaN layer has a narrower bandgap than the $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.(1-x)}}\text{N}$ layer, which may be referred to as a barrier layer because of its wider bandgap. Due to the bandgap mismatch, large conduction-band offset, and spontaneous and piezoelectric polarization properties of the group-III nitride layers, highly-mobile 2DEG may be generated in the GaN layer (having the narrower bandgap) near the interface of the heterostructure to form a conductive channel in the GaN layer (which is referred to as the channel layer). Compared to silicon-based transistors, GaN-based transistors devices generally have high breakdown field, high electron mobility, low resistance, high current, faster-switching speed, high thermal conductivity, and excellent reverse-recovery performance, and thus may be suitable for applications where a low-loss and high-efficiency performance may be desired, such as power electronics.

[0051] A GaN-based transistor may include a gate structure positioned between a source structure and a drain structure. The drain structure may include a metal contact that is coupled to the channel

layer and forms an ohmic contact with the channel layer. The source structure may include a metal contact that is coupled to the channel layer and forms an ohmic contact with the channel layer. Depending on the architecture of the gate structure, a GaN-based transistor may be an enhancement mode high electron mobility transistor (e-HEMT) or a depletion mode high electron mobility transistors (d-HEMT). For example, the gate structure of an e-HEMT may include a p-GaN layer formed over the barrier layer and a gate electrical contact (a metal electrode) formed on the p-GaN layer, which together form a p-GaN gate structure. The p-GaN layer may be doped using, for example, magnesium (Mg), which is an acceptor that makes the GaN layer p-type or p-doped. This p-GaN layer depletes electrons in the 2DEG channel under the p-GaN gate structure, such that the path between the source and gate may be disabled when no gate drive voltage is applied to the gate electrical contact. When a positive voltage above the gate threshold voltage is applied to the gate electrical contact, the gate structure may attract electrons to replete the 2DEG under the gate structure, thereby turning on the e-HEMT. In contrast, the gate structure of a d-HEMT may include an insulator layer (e.g., a dielectric layer) over the barrier layer and a gate electrical contact (e.g., a metal electrode) on the insulator layer. When no voltage signal is applied to the gate electrical contact, the 2DEG under the gate structure may not be depleted such that the path in the channel layer between the drain structure and the source structure may be enabled even without a positive gate voltage. A d-HEMT can be turned off by applying a negative threshold voltage to the gate electrical contact to deplete electrons from the 2DEG under the gate structure. In some applications such as switch-mode power applications, e-HEMTs, rather than d-HEMTs, may be used in order to, for example, decrease leakage current, reduce power loss, simplify the driving circuit, and/or improve device stability.

[0052] FIG. 1 is a cross-sectional view of an example of an enhancement mode HEMT **100** with an overlaid circuit model of HEMT **100**. In the illustrated example, HEMT **100** includes a channel layer **102**, a barrier layer **104** over channel layer **102**, and a gate semiconductor layer **106** over barrier layer **104**. Channel layer **102** and barrier layer **104** have different energy bandgaps. For example, channel layer **102** may include undoped GaN, and barrier layer **104** may include undoped AlGaN, which has a wider bandgap than GaN. The configuration of barrier layer **104** over channel layer **102** can induce a two-dimensional electron gas in channel layer **102** near the interface between channel layer **102** and barrier layer **104**. A gate electrical contact **112** on gate semiconductor layer **106** forms a gate structure with gate semiconductor layer **106**. Gate semiconductor layer **106** may be a p-doped semiconductor layer (e.g., a p-GaN layer) and can deplete electrons in the 2DEG under gate semiconductor layer **106**. Gate electrical contact **112** and gate semiconductor layer **106** may form a junction having an energy barrier height that depends on the energy band structures of the materials of gate electrical contact **112** and gate semiconductor layer **106**. In the illustrated example, gate electrical contact **112** and gate semiconductor layer **106** form a Schottky junction. A source electrical contact **114** and a drain electrical contact **116** are electrically coupled to channel layer **102** on opposing sides of gate semiconductor layer **106**. A dielectric layer **118** is deposited on barrier layer **104**, gate semiconductor layer **106**, and gate electrical contact **112** to encapsulate and isolate the conductive materials. Source electrical contact **114** and drain electrical contact **116** pass through dielectric layer **118** and barrier layer **104** to contact channel layer **102**.

[0053] FIG. 1 also shows a circuit model of HEMT **100**. In the circuit model, a gate contact node **120** is in or on gate electrical contact **112**, and a gate semiconductor layer node **122** is in gate semiconductor layer **106**. A source-side channel node **124** and a drain-side channel node **126** are at an interface between channel layer **102** and barrier layer **104**. A source contact node **128** is at the interface between source electrical contact **114** and channel layer **102**, while a drain contact node **130** is at the interface between drain electrical contact **116** and channel layer **102**.

[0054] At the gate structure, a Schottky diode (D.sub.SH) **140** and a Schottky capacitor (C.sub.SH) **142** are formed by gate electrical contact **112** and gate semiconductor layer **106**, and are electrically

coupled between gate contact node **120** and gate semiconductor layer node **122**. The cathode terminal of Schottky diode **140** is electrically coupled to the gate contact node **120**, and the anode terminal of Schottky diode **140** is electrically coupled to the gate semiconductor layer node **122**. Similarly, a first terminal of Schottky capacitor **142** is electrically coupled to gate contact node **120**, and a second terminal of Schottky capacitor **142** is electrically coupled to gate semiconductor layer node **122**.

[0055] At the source side of a region under the gate structure, a p-type, intrinsic, n-type (p-i-n) source-side diode (D.sub.s) **144** and a gate-to-source capacitor (C.sub.gs) **146** are formed by gate semiconductor layer **106**, barrier layer **104**, and channel layer **102**. Source-side diode **144** and gate-to-source capacitor **146** are electrically coupled between gate semiconductor layer node **122** and source-side channel node **124**. The anode terminal of source-side diode **144** is electrically coupled to gate semiconductor layer node **122**, and the cathode terminal of source-side diode **144** is electrically coupled to source-side channel node **124**. A first terminal of gate-to-source capacitor **146** is electrically coupled to gate semiconductor layer node **122**, and a second terminal of gate-to-source capacitor **146** is electrically coupled to source-side channel node **124**.

[0056] At the drain side of the region under the gate structure, a p-i-n drain-side diode (D.sub.d) **148** and a gate-to-drain capacitor (C.sub.gd) **150** are formed by gate semiconductor layer **106**, barrier layer **104**, and channel layer **102**. Drain-side diode **148** and gate-to-drain capacitor **150** are electrically coupled between gate semiconductor layer node **122** and drain-side channel node **126**. The anode terminal of drain-side diode **148** is electrically coupled to gate semiconductor layer node **122**, and the cathode terminal of drain-side diode **148** is electrically coupled to drain-side channel node **126**. A first terminal of gate-to-drain capacitor **150** is electrically coupled to gate semiconductor layer node **122**, and a second terminal of gate-to-drain capacitor **150** is electrically coupled to drain-side channel node **126**.

[0057] The circuit model of HEMT **100** includes a field effect transistor (FET) **152** having a gate terminal electrically coupled to gate semiconductor layer node **122**. The source terminal of FET **152** is electrically coupled to source-side channel node **124**, and the drain terminal of FET **152** is electrically coupled to drain-side channel node **126**. The circuit model of HEMT **100** also includes a FET **154**, where the channel of FET **154** and a channel-to-source resistor (R.sub.cs) **156** are serially connected between source-side channel node **124** and source contact node **128**. The circuit model of HEMT **100** further includes a FET **158**, where the channel of FET **158** and a channel-to-drain resistor (R.sub.cd) **160** are serially connected between drain-side channel node **126** and drain contact node **130**.

[0058] Having the Schottky junction between gate electrical contact **112** and gate semiconductor layer **106** (shown by Schottky diode **140** and Schottky capacitor **142** coupled between gate contact node **120** and gate semiconductor layer node **122**) may result in low gate leakage current during operations of HEMT **100**. When a positive voltage is applied to gate electrical contact **112** (e.g., gate contact node **120**), the positive voltage may reverse bias the Schottky junction, which can result in a low gate leakage current. However, as illustrated by the circuit model, HEMT **100** with the gate structure having a Schottky junction between gate electrical contact **112** and gate semiconductor layer **106** includes a back-to-back diode configuration at gate semiconductor layer node **122**, where the anode terminals of Schottky diode **140**, source-side diode **144**, and drain-side diode **148** are electrically coupled to a same node (gate semiconductor layer node **122**). Such a back-to-back diode configuration may result in a voltage divider and a floating, quasi-neutral node (e.g., gate semiconductor layer node **122**) in gate semiconductor layer **106** under some circumstances.

[0059] For example, under direct current (DC) conditions, the floating, quasi-neutral node in gate semiconductor layer **106** is in a voltage divider formed by Schottky diode **140** and source-side diode **144** or drain-side diode **148**. The voltage divider may divide the voltage level applied to gate electrical contact **112**, such that the effective voltage level applied to gate semiconductor layer **106**

is lower than the voltage level applied to gate electrical contact **112**. As a result, the drain current ($I_{sub.d}$) may be lower, and the drain-to-source on-state resistance ($R_{sub.DSON}$) may be higher in HEMT **100** with Schottky diode **140** than in an HEMT without Schottky diode **140**.

[0060] In switching events, the floating, quasi-neutral node in gate semiconductor layer **106** may be biased by past switching events due to its floating nature. This biasing may result in non-deterministic behavior of HEMT **100** and may adversely affect the drain current of HEMT **100**. Furthermore, when HEMT **100** is switched on, Schottky capacitor **142** may form a low impedance path initially, which may quickly pull the voltage level of gate semiconductor layer node **122** up to the voltage applied to gate contact node **120**. However, following the initial switching, the voltage level of gate semiconductor layer node **122** may settle to a voltage level proportional to the voltage level applied on gate electrical contact **112** due to voltage dividing by the voltage divider. This voltage change at gate semiconductor layer node **122** may result in a drain current that has an initial peak and subsequently decreases and settles. Therefore, HEMT **100** may experience a dynamic gate overdrive voltage ($V_{sub.GT}$).

[0061] In some examples, to address these effects, an HEMT may include a gate electrical contact that includes a first metal portion and a second metal portion. The first metal portion forms a first junction with a first semiconductor portion of the gate semiconductor layer (e.g., gate semiconductor layer **106**), whereas the second metal portion forms a second junction with a second semiconductor portion of the gate semiconductor layer. An energy barrier height of the first junction is greater than the energy barrier height of the second junction. For example, the first junction may be a high-barrier Schottky junction, and the second junction may be a low-barrier junction such as a low-barrier Schottky (LBS) junction or ohmic junction.

[0062] FIG. 2 is a cross-sectional view of another example of an enhancement mode HEMT **200** with an overlaid circuit model of HEMT **200**. In the illustrated example, HEMT **200** includes most components of HEMT **100**, and has a gate electrical contact **212** that is different from gate electrical contact **112** of HEMT **100**. Gate electrical contact **212** includes a first metal portion **214** and a second metal portion **216** that are electrically shorted and have the same voltage level. First metal portion **214** and a portion of gate semiconductor layer **106** form a first junction, such as a Schottky junction described above. Second metal portion **216** and another portion of the gate semiconductor layer **106** form a second junction. In the example shown in FIG. 2, the second junction is an ohmic junction formed between second metal portion **216** and gate semiconductor layer **106**. In another example, the second junction may be a Schottky junction having an energy barrier height lower than the energy barrier height of the first junction.

[0063] As shown in FIG. 2, the circuit model of HEMT **200** includes an ohmic resistor ($R_{sub.OH}$) **218** electrically coupled between gate contact node **120** and gate semiconductor layer node **122**. Ohmic resistor **218** is formed by the junction between second metal portion **216** and the second semiconductor portion of gate semiconductor layer **106**. The ohmic junction or another low-barrier junction may provide an electrical path through which gate semiconductor layer node **122** (e.g., gate semiconductor layer **106**) may be biased by the voltage applied to gate electrical contact **212**, and thus may reduce the voltage drop between gate electrical contact **212** and gate semiconductor layer node **122**. In addition, the small (e.g., close to zero) voltage drop can be the same under DC conditions and in switching events. Accordingly, the effective gate drive voltage can be increased (under DC conditions) and stabilized (in switching events), and thus the drain current (I_d) can also be increased. In addition, because gate semiconductor layer **106** can be driven to a voltage level that more accurately tracks the voltage level at gate electrical contact **112** by bypassing Schottky capacitor **142**, the drain current may be more stable in switching events, rather than having an initial peak and subsequently decreasing as described above with respect to FIG. 1. The surface areas of the first junction and the second junction may be scaled to obtain a target gate leakage while obtaining the stabilized gate overdrive voltage ($V_{sub.GT}$) and increased drain current ($I_{sub.d}$). For example, by having a surface area of the second junction relatively small compared to

the surface area of the first junction, gate leakage may remain low, and a path to bias gate semiconductor layer **106** may still be provided to achieve stabilized gate overdrive voltage ($V_{sub.GT}$) and increased drain current ($I_{sub.d}$).

[0064] GaN-based HEMTs described above are attractive for high frequency and high power applications due to, for example, high breakdown voltage, high electron mobility, low static resistance, and high thermal conductivity of GaN-based HEMTs. However, GaN-based HEMTs may suffer from current collapse, which is an undesirable phenomenon of an increase in the dynamic on-state drain-to-source resistance (dR_{on}) that occurs during high-voltage switching operations, for example, when the GaN-based HEMT is turned on and off and a high voltage level is applied to the drain of the transistor. Current collapse may result in, for example, power loss, temperature increase, and reliability issues.

[0065] Current collapse may be caused by electron and/or hole trapping, and can appear as a transient and recoverable reduction in the drain current (and increase in the effective resistance) during high-voltage switching operations. Several factors may contribute to current collapse, including electrons trapped at the surface of the barrier layer and/or in the buffer layer in the off state with a high drain bias, and hot electrons trapped during high voltage switching. For example, for an HEMT including an AlGaIn barrier layer and a GaN channel layer, when the HEMT is in the off state and has the following voltage states: a high drain voltage (e.g., a few hundred volts such as about 600V or higher), a gate voltage below the threshold, and a grounded source voltage, high-energy electrons from the 2DEG may be injected towards the top of the AlGaIn barrier layer and trapped by the surface states (resulting in a negatively charged surface) due to the high electric field caused by large negative gate-to-drain bias. The large positive drain-to-substrate voltage may cause electrons to be injected from the substrate and trapped in the buffer stack between the channel layer and the substrate. Furthermore, in the off state, the high drain voltage induces a high electric field, which may ionize holes in the heterostructure between the gate and the drain contact structures or underneath the drain contact structure. These ionized holes may be pulled towards the gate and/or source contact structures in the off state due to the electric field caused by the biasing, which may leave fixed negative charges in the heterostructure. When the HEMT operates with high drain voltages (e.g., a few hundred volts), a hard switching can generate hot electrons in the channel due to the high electric field, and the hot electrons may be injected and trapped, for example, in the dielectric, at the surface of the barrier layer, in the channel layer, and/or in the buffer layer. The trapped negative charges and the fixed negative charges in the heterostructure may at least partially deplete carriers (e.g., electrons) in the 2DEG channel, causing a reduction in the channel carrier density and thus an increase in the resistance.

[0066] When the HEMT is switched on (e.g., when the gate voltage is above the threshold voltage in an e-HEMT), the dynamic on-state drain-to-source resistance may be high initially as compared to a static on-state drain-to-source resistance, due to the lower channel carrier density caused by carrier depletion by the negative charges. The trapped electrons and fixed negative charges may be slowly detrapped or neutralized by holes injected from the drain, and the remaining negative charges may continue to deplete the 2DEG, leading to a slow reduction of the drain-to-source resistance.

[0067] In some examples, a hybrid drain contact may be used to mitigate the electron trapping and current collapse effects. For example, a hybrid drain contact structure may include, in addition to the drain contact structure that forms an ohmic contact with the channel layer, a p-type drain contact structure that includes a p-doped semiconductor layer (e.g., a p-GaN layer) positioned below a drain electrical contact (e.g., a metal layer). Therefore, the hybrid drain contact structure includes both the drain contact structure and the p-type drain contact structure that are electrically coupled and driven by the same drain voltage. The hybrid drain contact structure may be used to inject holes to neutralize the trapped electrons at the off state and during the switching, due to the high voltage applied to the drain contact, thereby reducing the current collapse effect.

[0068] However, in an enhancement mode HEMT with the p-type drain contact structure, in order to avoid depleting the 2DEG channel in the drain region by the p-doped semiconductor layer (e.g., a p-GaN layer), the barrier layer (e.g., an AlGaN layer) under the p-doped semiconductor layer of the p-type drain contact structure may need to be thicker than the barrier layer under the p-doped semiconductor layer at the gate of the enhancement mode HEMT, and/or the p-doped semiconductor layer of the p-type drain contact structure may need to be thinner than the p-doped semiconductor layer at the gate of the enhancement mode HEMT. The different thicknesses are needed for normally-off operation because the barrier layer below the p-type gate contact structure may need to be thin enough to ensure that the threshold voltage (V_t) is greater than zero (positive), while the barrier layer below the p-type drain contact structure needs to be thick to prevent the p-type drain contact structure from depleting the electrons from the 2DEG channel below it and enable current flow during the on state. In this way, in the off state, the 2DEG is only depleted at the gate of the e-HEMT with the p-type drain contact structure. Having a thicker p-doped semiconductor layer at the p-type gate contact structure and a thinner p-doped semiconductor layer at the p-type drain contact structure may achieve similar results. Therefore, fabricating normally-off GaN-based transistors (e.g., e-HEMTs) with p-type drain contact structures may include growing the barrier layer or the p-doped semiconductor layer twice (or selective etching) to attain different barrier layer thicknesses and/or different thicknesses of the p-doped semiconductor layer at the gate and the drain. The additional processing steps (e.g., epitaxial growth) may increase manufacturing costs and reduce both yield and throughput.

[0069] In some enhancement mode GaN-based HEMTs, multiple p-type drain contact structures may be used to inject holes to neutralize the trapped electrons, thereby mitigating the current collapses. The barrier layer under the p-GaN layer of the p-type drain contact structures may have the same thickness as the barrier layer under the p-GaN layer at the gate of the enhancement mode HEMT, such that one epitaxial growth process may be used to grow the barrier layer to reduce the manufacturing cost. The p-GaN layer of the p-type drain contact structures and the p-GaN layer of the gate structure may also be fabricated using the same processes and may have the same thickness. Each of the multiple p-type drain contact structures may be physically separate from the remaining p-type drain contact structures, such that the 2DEG channel may only be depleted at isolated regions at the drain, whereas other regions of the 2DEG channel at the drain are not depleted and thus may allow current to pass through. In this way, the current collapse may be reduced due to the hole injection by the multiple p-type drain contact structures, and the fabrication cost may also be reduced.

[0070] FIG. 3A includes top-down views of an example of an HEMT **300** including a hybrid drain contact structure. In the illustrated example, HEMT **300** is an enhancement mode GaN-based transistor that includes multiple p-type drain contact structures, where each p-type drain contact structure of the multiple p-type drain contact structures includes a p-doped semiconductor layer (e.g., a p-GaN block) and a drain contact structure (e.g., a metal contact). The top-down views include a top-down view **302** of HEMT **300** that includes some contact structures (e.g., metal contact layers) at the drain, gate, and source of HEMT **300**, and a top-down view **322** of HEMT **300** that does not include some contact structures at the drain, gate, and source of HEMT **300**. Top-down view **302** depicts a source contact structure **310**, a gate contact structure **309**, a drain contact structure **304**, and a barrier layer **330**. Top-down view **322** depicts the layers of HEMT **300** underneath the contact structures depicted in the top-down view **302**, including a channel layer **348**, a p-GaN gate structure **328** that enables HEMT **300** to operate in an enhancement mode, barrier layer **330**, and an array of p-GaN blocks **323**, **324**, **325**, and **326**. The array of p-GaN blocks **323**, **324**, **325**, and **326** on barrier layer **330** and p-GaN gate structure **328** may be fabricated by epitaxial growth of a p-GaN layer on barrier layer **330** and patterning the p-GaN layer by selective etching. In one example, barrier layer **330** includes an AlGaN layer, and channel layer **348** includes a GaN layer, where the AlGaN layer and the GaN layer form a heterostructure that induces 2DEG in the

GaN layer at the interface between the AlGaN layer and the GaN layer.

[0071] In HEMT **300**, source contact structure **310** may be formed on and in contact with a source region of channel layer **348**, to form an ohmic source contact. Gate contact structure **309** is positioned on p-GaN gate structure **328** to form the gate structure of the enhancement mode HEMT. Drain contact structure **304** may be formed on and in contact with a drain region of channel layer **348** to form an ohmic drain contact with the channel layer. Drain contact structure **304** may also be formed on and in contact with the array of p-GaN blocks **323**, **324**, **325**, and **326**, to form multiple p-type drain contact structures for mitigating current collapse. The ohmic drain contact and the multiple p-type drain contact structures form a hybrid drain contact structure. In the illustrated example, the array of p-GaN blocks **323**, **324**, **325**, and **326** are underneath drain contact structure **304**. In some other examples, the array of p-GaN blocks **323**, **324**, **325**, and **326** may be positioned on a side of drain contact structure **304** in the x direction (the channel length direction of HEMT **300**). In the illustrated example, each p-GaN block of the array of p-GaN blocks **323**, **324**, **325**, and **326** has a rectangular shape in the top-down view. In other examples, each p-GaN block of the array of p-GaN blocks **323**, **324**, **325**, and **326** may have another regular, irregular, or arbitrary shape.

[0072] As shown in top-down view **322**, p-GaN blocks **323**, **324**, **325**, and **326** have widths **T1**, **T2**, **T3**, and **T4**, respectively, along the channel width direction (e.g., y direction). In the illustrated example, p-GaN blocks **323**, **324**, **325**, and **326** have the same widths. In other examples, widths **T1**, **T2**, **T3**, and **T4** may be unequal. Top-down view **322** also shows that p-GaN blocks **323**, **324**, **325**, and **326** have lengths **T1'**, **T2'**, **T3'**, and **T4'**, respectively, in the channel length direction (e.g., the x direction). In the example illustrated in FIG. 3A, p-GaN blocks **323**, **324**, **325**, and **326** have equal lengths. In other examples, lengths **T1'**, **T2'**, **T3'**, and **T4'** may be unequal. In the illustrated example, the length of drain contact structure **304** may be similar to the lengths of p-GaN blocks **323**, **324**, **325**, and **326** in the channel length direction. In other examples, the length of drain contact structure **304** may be different from (e.g., shorter or longer than) the lengths of p-GaN blocks **323**, **324**, **325**, and **326** in the channel length direction.

[0073] As also shown in top-down view **322** of FIG. 3A, regions **347** of HEMT **300** do not have p-GaN gate structure **328** or the array of p-GaN blocks **323**, **324**, **325**, and **326** over barrier layer **330**. Therefore, the 2DEG in regions **347** is not depleted, and there are conductive paths in the channel layer between the gate and the drain of HEMT **300** even when HEMT **300** is in the off state. For example, the conductive path of the channel layer between p-GaN blocks **323** and **324** may have a width **D1**, the conductive path of the channel layer between p-GaN blocks **324** and **325** may have a width **D2**, and the conductive path of the channel layer between p-GaN blocks **325** and **326** may have a width **D3**. **D1**, **D2**, and **D3** may be the same or may be different. As such, barrier layer **330** under p-GaN gate structure **328** and the array of p-GaN blocks **323**, **324**, **325**, and **326** can have the same thickness, without turning off the 2DEG channel between the gate and the drain of HEMT **300** even when HEMT **300** is in the off state. In this way, each of barrier layer **330** and the p-GaN layer above barrier layer **330** can be grown in one epitaxial growth process.

[0074] FIG. 3B includes top-down views of another example of an HEMT **300'** including a hybrid drain contact structure. HEMT **300'** may be similar to HEMT **300**, but may include p-GaN blocks having shapes different from the shapes of p-GaN blocks **323**, **324**, **325**, and **326** of HEMT **300**. As HEMT **300**, HEMT **300'** is an enhancement mode GaN-based transistor that includes multiple p-type drain contact structures, where each p-type drain contact structure of the multiple p-type drain contact structures includes a p-doped semiconductor layer (e.g., a p-GaN block) and a drain contact structure (e.g., a metal contact). The top-down views include a top-down view **302'** of HEMT **300'** that shows some contact structures (e.g., metal contact layers) at the drain, gate, and source of HEMT **300'**, and a top-down view **322'** of HEMT **300'** that does not include some contact structures at the drain, gate, and source of HEMT **300'**. Top-down view **302'** depicts source contact structure **310**, gate contact structure **309**, drain contact structure **304**, and barrier layer **330**. Top-

down view **322'** depicts the layers of HEMT **300'** underneath the contact structures depicted in top-down view **302'**, including channel layer **348**, p-GaN gate structure **328** that enables HEMT **300'** to function in a normally-off mode, barrier layer **330**, and an array of p-GaN blocks **323'**, **324'**, **325'**, and **326'**. The array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** on barrier layer **330** and p-GaN gate structure **328** may be formed, for example, by epitaxial growth of a p-GaN layer over barrier layer **330** and patterning the p-GaN layer by selective etching, or by selective area growth.

[0075] In HEMT **300'**, source contact structure **310** may be formed on and in contact with a source region of channel layer **348**, to form an ohmic source contact. Gate contact structure **309** is positioned on p-GaN gate structure **328** to form the gate structure of the enhancement mode HEMT. Drain contact structure **304** may be formed on and in contact with a drain region of channel layer **348** to form an ohmic drain contact with the channel layer. Drain contact structure **304** may also be formed on and in contact with the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'**, to form multiple p-type drain contact structures for mitigating current collapse. The ohmic drain contact and the multiple p-type drain contact structures form a hybrid drain contact structure.

[0076] In the illustrated example, the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** are underneath drain contact structure **304**, where each p-GaN block of the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** has a hexagonal shape in the top-down view. In other examples, edges of each p-GaN block of the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** may be straight lines or may include curves (e.g., a parabolic curve), and each p-GaN block may have a regular, irregular, or arbitrary shape. In the illustrated example, each p-GaN block is tapered in a direction towards p-GaN gate structure **328**. The tapering may be linear or nonlinear (e.g., parabolic). As shown in top-down view **322'**, p-GaN blocks **323'**, **324'**, **325'**, and **326'** have lengths T5, T6, T7, and T8, respectively, in the channel length direction (e.g., the x direction). In the example illustrated in FIG. 3B, p-GaN blocks **323'**, **324'**, **325'**, and **326'** have equal lengths. In other examples, lengths T5, T6, T7, and T8 may be unequal. In the illustrated example, the length of drain contact structure **304** may be similar to the lengths of p-GaN blocks **323'**, **324'**, **325'**, and **326'** in the channel length direction. In other examples, the length of drain contact structure **304** may be different from the lengths of p-GaN blocks **323**, **324**, **325**, and **326** in the channel length direction.

[0077] As also shown in top-down view **322'** of FIG. 3A, regions **347** of HEMT **300'** do not have p-GaN gate structure **328** or the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** over barrier layer **330**. Therefore, the 2DEG in regions **347** is not depleted, and there are conductive paths in the channel layer between the gate and the drain of HEMT **300'** even when HEMT **300'** is in the off state. For example, the distance between the centers of p-GaN blocks **323'** and **324'** is D4, which may be much larger than the width of p-GaN block **323'** or **324'** in the channel width direction (e.g., y direction). The distance between the centers of p-GaN blocks **324'** and **325'** is D5, which may be much larger than the width of p-GaN block **324'** or **325'** in the channel width direction. The distance between the centers of p-GaN blocks **325'** and **326'** is D6, which may be much larger than the width of p-GaN block **324'** or **325'** in the channel width direction. As such, barrier layer **330** under p-GaN gate structure **328** and the array of p-GaN blocks **323'**, **324'**, **325'**, and **326'** can have the same thickness, without turning off the 2DEG channel between the gate and the drain of HEMT **300'** even when HEMT **300'** is in the off state. In this way, each of barrier layer **330** and the p-GaN layer above barrier layer **330** can be grown in one epitaxial growth process. In addition, the linear or nonlinear tapering of p-GaN blocks **323'**, **324'**, **325'**, and **326'**, and thus the linear or nonlinear tapering of 2DEG channels between the p-GaN blocks near the drain, may facilitate the movement of the carriers in the 2DEG channel, such that the channel resistance of HEMT **300'** may be lower than the channel resistance of HEMT **300** having the rectangular-shaped p-GaN blocks.

[0078] FIG. 3C is a cross-sectional view **342** of an example of a cross-section of the HEMT of FIG. 3A or 3B along a line **303**. Cross-sectional view **342** shows drain contact structure **304** that is in ohmic contact with a channel layer **343** of HEMT **300** or **300'**. In the illustrated example, the HEMT includes a substrate **363**, channel layer **343** grown on substrate **363**, and barrier layer **330**

grown on channel layer **343**. Substrate **363** may include, for example, silicon, silicon carbide, sapphire, gallium nitride, or other suitable substrate material or substrate consisting of multiple materials. In some examples, the substrate material (e.g., silicon) may have a large lattice structure mismatch with GaN, and substrate **363** may include one or more seed layers or buffer layers (e.g., aluminum nitride, not shown in FIG. 3C) deposited on it to facilitate high quality epitaxial growth of the GaN-based material layers (e.g., with low defect density and appropriate thicknesses).

[0079] Channel layer **343** may include, for example, GaN, which may be doped or undoped. Barrier layer **330** may include, for example, $\text{Al}_{\text{sub}}\text{xGa}_{\text{sub}}(1-\text{x})\text{N}$. In some examples, the thickness of barrier layer **330** may be in the range of a few nanometers (e.g., about 1 nm) to tens of microns (e.g., about 20 microns). In some examples, the thickness of barrier layer **330** can be in the range of a few nanometers (e.g., about 1 nm) to hundreds of nanometers (e.g., about 1000 nm). In some examples, the $\text{Al}_{\text{sub}}\text{xGa}_{\text{sub}}(1-\text{x})\text{N}$ material in barrier layer **330** may include other elements, such as indium (In). For example, barrier layer **330** may have a composition $\text{Al}_{\text{sub}}(\text{X})\text{In}_{\text{sub}}(\text{Y})\text{Ga}_{\text{sub}}(1-\text{X}-\text{Y})\text{N}$, where X and Y are the concentrations of Aluminum and Indium, respectively, and may be selected to achieve the desired energy band structure and lattice constant. In some examples, barrier layer **330** may have a uniform composition, such as a uniform composition of $\text{Al}_{\text{sub}}0.3\text{Ga}_{\text{sub}}0.7\text{N}$. In other examples, barrier layer **330** may have a graded composition of Aluminum, Gallium, and Indium, in different regions of a same layer or in different sublayers. Due to the different energy band structures of the materials of channel layer **343** and barrier layer **330**, channel layer **343** and barrier layer **330** form a heterostructure that induces 2DEG **364** in channel layer **343** near the interface between channel layer **343** and barrier layer **330**.

[0080] To form the gate structure of an e-HEMT, a p-doped semiconductor layer (e.g., p-GaN layer) may be grown on a top side **331** of barrier layer **330** and patterned by selective etching (e.g., using an etch mask) to form p-GaN gate structure **328**, and then gate contact structure **309** may be deposited on p-GaN gate structure **328**. The thicknesses of p-GaN gate structure **328** and barrier layer **330** under p-GaN gate structure **328** may be selected such that 2DEG **364** under p-GaN gate structure **328** may be depleted in depleted region DR1, such that the HEMT may be off without a positive gate voltage. In one example, gate contact structure **309** and p-GaN gate structure **328** may form a Schottky junction as described above with respect to, for example, FIG. 1.

[0081] Barrier layer **330** may be etched to expose regions of channel layer **343**, and drain contact structure **304** and source contact structure **310** may be deposited on the exposed regions of channel layer **343** to form ohmic drain and source contacts with channel layer **343**. The ohmic contact is a low resistance junction that provides a low-resistance conduction between channel layer **343** and either source contact structure **310** or drain contact structure **304**. In some examples, source contact structure **310** and drain contact structure **304** may be fabricated such that they extend into barrier layer **330** and couple to channel layer **343** through quantum tunneling, rather than ohmic contact. As illustrated, in the cross-section shown in cross-sectional view **342**, 2DEG **364** is not deleted near the drain region. As shown in FIGS. 3A and 3B, in HEMTs **300** and **300'**, there are multiple regions between the p-GaN blocks at the drain region where 2DEG **364** is not deleted in the off state and can provide a low-resistance current path.

[0082] FIG. 3D is a cross-sectional view **362** of another cross-section of the HEMT of FIG. 3A or 3B along a line **333**. Cross-sectional view **362** shows drain contact structure **304** on a p-GaN block **325** (or p-GaN block **325'**). As described above, in some examples, p-GaN block **325** may be fabricated using the same processes for fabricating p-GaN gate structure **328**, and thus may have the same material and thickness as p-GaN gate structure **328**. The region of barrier layer **330** under p-GaN block **325** and the region of barrier layer **330** under p-GaN gate structure **328** may also be fabricated using the same processes and may have the same thickness. As such, 2DEG **364** under p-GaN gate structure **328** may be depleted in depleted region DR2, and 2DEG **364** under p-GaN block **325** may also be depleted in depleted region DR3. Other p-type drain contact structures formed by other p-GaN blocks (e.g., p-GaN blocks **323**, **324**, and **326**) may have similar

configurations. As the p-GaN gate structure described above with respect to FIG. 1, drain contact structure **304** and the p-GaN blocks (e.g., p-GaN blocks **323**, **324**, **325**, and **326**) may form Schottky junctions to inject holes for neutralizing trapped electrons. Because each of the multiple p-type drain contact structures is physically separate from the remaining p-type drain contact structures, 2DEG **364** may only be depleted at isolated regions at the drain, whereas other regions of 2DEG **364** at the drain are not depleted and thus may provide a low-resistance current path. In this way, the current collapse may be reduced, and the fabrication cost may also be reduced.

[0083] Since 2DEG **364** under the p-GaN blocks at the drain region may be depleted by the p-GaN blocks in the static on state, the effective channel width of the HEMT may be reduced and thus the static on-state resistance of HEMT **300** or **300'** may be higher than the static on-state resistance of an HEMT without the p-GaN blocks at the drain region. Therefore, it may be desirable to reduce the sizes (e.g., widths) of the p-GaN blocks to reduce the static on-state resistance of HEMT **300** or **300'**. On the other hand, increasing the sizes of the p-GaN block may improve the hole injection efficiency and reduce the dynamic on-state resistance of the HEMT. Therefore, the sizes (e.g., widths) of the p-GaN blocks may need to be selected to balance the reduction of the dynamic on-state resistance and the increase of the static on-state resistance.

[0084] In addition, as described above with respect to FIG. 1, the Schottky junction in the HEMT may result in a back-to-back diode configuration, which in turn may result in a floating, quasi-neutral node in the p-GaN layer. The back-to-back diodes may form a voltage divider, which may reduce the effective drive voltage, reduce drain current ($I_{sub,d}$), and increase drain-to-source on-state resistance ($R_{sub,DS(on)}$). The reduction in the effective drain drive voltage may result in a decrease in the hole injection capability of the p-type drain contact structure, and thus the p-type drain contact structure may be less effective in mitigating the current collapse. In switching events, the floating, quasi-neutral node may be biased based on past switching events due to its floating nature, and thus may have a memory effect. This biasing that depends on past switching events may result in a non-deterministic behavior of the HEMT and may adversely affect the drain current of the HEMT. In addition, due to the reduced hole injection capability, the dynamic $R_{sub,DS(on)}$ may be higher and the switching time may be longer.

[0085] FIG. 3E is a cross-sectional view of an example of a hybrid drain contact structure including a Schottky contact, with an overlaid circuit model of the hybrid drain contact structure. The hybrid drain contact structure includes a portion of drain contact structure **304** that forms an ohmic contact with channel layer **343**, and a portion of drain contact structure **304** that forms a Schottky contact with p-GaN block **325**. As illustrated, p-GaN block **325** is formed on barrier layer **330**, which is grown on channel layer **343**. In the circuit model, a drain contact node **366** is in drain contact structure **304**, and a p-GaN drain node **374** is in p-GaN block **325**. One or more channel nodes, such as a first channel node **382** and a second channel node **388**, are at an interface between channel layer **343** and barrier layer **330**.

[0086] A Schottky diode **370** and a Schottky capacitor **372** are formed by drain contact structure **304** and p-GaN block **325**, and are electrically coupled between drain contact node **366** and p-GaN drain node **374**. The cathode terminal of Schottky diode **370** is electrically coupled to drain contact node **366**, and the anode terminal of Schottky diode **370** is electrically coupled to the p-GaN drain node **374**. Similarly, a first terminal of Schottky capacitor **372** is electrically coupled to drain contact node **366**, and a second terminal of Schottky capacitor **372** is electrically coupled to p-GaN drain node **374**.

[0087] A first side diode **378** (e.g., a p-i-n diode) and a first side capacitor **380** are formed by p-GaN block **325**, barrier layer **330**, and channel layer **343**. First side diode **378** and first side capacitor **380** are electrically coupled between p-GaN drain node **374** and first channel node **382**. The anode terminal of first side diode **378** is electrically coupled to p-GaN drain node **374**, and the cathode terminal of first side diode **378** is electrically coupled to first channel node **382**. A first terminal of first side capacitor **380** is electrically coupled to p-GaN drain node **374**, and a second

terminal of first side capacitor **380** is electrically coupled to first channel node **382**.

[0088] In examples where p-GaN block **325** is separate from the portion of drain contact structure **304** that forms the ohmic contact with channel layer **343**, a second side diode **384** (e.g., a p-i-n diode) and a second side capacitor **386** may be formed by p-GaN block **325**, barrier layer **330**, and channel layer **343**. Second side diode **384** and second side capacitor **386** are electrically coupled between p-GaN drain node **374** and second channel node **388**. The anode terminal of second side diode **384** is electrically coupled to p-GaN drain node **374**, and the cathode terminal of second side diode **384** is electrically coupled to second channel node **388**. A first terminal of second side capacitor **386** is electrically coupled to p-GaN drain node **374**, and a second terminal of second side capacitor **386** is electrically coupled to second channel node **388**. The circuit model of the hybrid drain contact structure also includes a field effect transistor (FET) **376** having a gate terminal electrically coupled to p-GaN drain node **374**. The source terminal of FET **376** is electrically coupled to first channel node **382**, and the drain terminal of FET **376** is electrically coupled to second channel node **388**.

[0089] As illustrated by the circuit model, the hybrid drain contact structure with the Schottky junction between drain contact structure **304** and p-GaN block **325** includes a back-to-back diode configuration at p-GaN drain node **374**, where the anode terminals of Schottky diode **370**, first side diode **378**, and second side diode **384** are electrically coupled to a same node (e.g., p-GaN drain node **374**). Such a back-to-back diode configuration may result in a voltage divider and a floating, quasi-neutral node (e.g., p-GaN drain node **374**) in p-GaN block **325** under some circumstances. In switching events, the floating, quasi-neutral node in p-GaN block **325** may be biased by past switching events due to its floating nature. This biasing may result in a non-deterministic behavior of the HEMT and may adversely affect the drain current of the HEMT. The floating, quasi-neutral node in p-GaN block **325** is in a voltage divider formed by Schottky diode **370** and first side diode **378** (and/or second side diode **384**). The voltage divider may divide the voltage level applied to drain contact structure **304**, such that the effective voltage level applied to p-GaN block **325** is lower than the voltage level applied to drain contact structure **304**, and thus the voltage applied across the first side diode **378** (and/or second side diode **384**) is reduced. As a result, the drive strength and the hole injection efficiency of the hybrid drain contact structure may be reduced, and the dynamic on-state resistance of the HEMT may still be high.

[0090] FIG. **4** includes a graph **400** illustrating reduction of the dynamic on-state resistance (dRon) of an HEMT using a hybrid drain contact. In graph **400**, the horizontal axis represents the drive voltage (e.g., VDD) applied to the drain or the voltage difference VDs between the drain and source of the HEMT before the HEMT is switched on, and the vertical axis represents normalized dynamic on-state resistance of the HEMT. The normalized dynamic on-state resistance is the ratio between the dynamic on-state resistance and the static on-state resistance. Plot **410** show normalized dynamic on-state resistance of HEMTs without hybrid drain contacts, such as HEMT **100** or **200** described above with respect to FIGS. **1** and **2**. Plots **420** show normalized dynamic on-state resistance of HEMTs having hybrid drain contacts, such as HEMT **300** or **300'** described above with respect to FIGS. **3A** and **3B**.

[0091] When V.sub.DS is low, for example, when the drive voltage applied to the drain is low, the dynamic on-state resistance of an HEMT without hybrid drain contacts (e.g., HEMT **100** or **200**) may be close to the static on-state resistance of the HEMT because the electron trapping and current collapse generally occur when the drive voltage applied to the drain (and thus VDs) is high as described above. Therefore, the normalized dynamic on-state resistance may be close to 1 at low drive voltage as shown by, for example, a data point **402** in FIG. **4**. As the drive voltage applied to the drain (and thus V.sub.DS) before a switching increases, more and more electrons may be trapped before and during the switching to deplete the 2DEG channel, and thus the current collapse may become more and more severe and the normalized dynamic on-state resistance may become larger. When the drive voltage applied to the drain (and thus V.sub.DS) before the switching is

sufficiently high (e.g., higher than the $V_{\text{sub.DS}}$ at a data point **404**), the 2DEG between the drain and the gate of the HEMT may be depleted such that the 2DEG channel is switched off, and thus the normalized dynamic on-state resistance may become very large.

[0092] When a hybrid drain contact in, for example, FIGS. **3A** and **3B**, is used in an HEMT (e.g., HEMT **300** or **300'**), if the drive voltage applied to the drain (and thus $V_{\text{sub.DS}}$) before switching is low, the electron trapping and current collapse may increase as the drive voltage applied to the drain increases, but if the voltage difference between the p-type drain contact structure and the channel layer under the p-type drain contact structure is not large enough to turn on the diodes, the p-type drain contact structure may inject very few holes, or does not inject holes at all, to neutralize the trapped negative charges. Therefore, the normalized dynamic on-state resistance may increase as the drive voltage applied to the drain increases. When the drive voltage applied to the drain (and thus $V_{\text{sub.DS}}$) before a switching is sufficiently high (e.g., higher than the VDs at a data point **404**), the voltage difference between the drain contact structure (e.g., drain contact structure **304**) and the channel layer under the p-GaN block can be sufficiently high to turn on the diodes (e.g., the p-i-n diode) to inject holes to neutralize the trapped negative charges. As a result, the normalized dynamic on-state resistance may reduce as shown by curves **420**, rather than becoming very large as shown by curves **410**. But the normalized dynamic on-state resistance may increase as the drain drive voltage continues to increase as shown by curves **420**.

[0093] As shown in FIG. **4**, using a hybrid drain contact structure in FIGS. **3A** and **3B**, the current collapse may be reduced but the normalized dynamic on-state resistance may still be significantly higher than 1 at high drain drive voltages. The high normalized dynamic on-state resistance may be at least partially caused by the back-to-back diode configuration of the hybrid drain contact structure with the Schottky contact. As described above, the back-to-back diode configuration may function as a voltage divider that effectively reduces the voltage level at the p-GaN block, and thus the efficiency of hole injection from the p-GaN block to the barrier layer and channel layer may be reduced.

[0094] According to certain examples, a drain contact structure disclosed herein includes a hybrid drain contact structure that includes both an ohmic contact coupled to the channel layer and a p-type drain contact structure that includes two or more junctions with different energy barrier heights formed by one or more metal materials and a p-type semiconductor layer (e.g., a p-GaN layer) over the barrier layer. In one example, the p-type drain contact structure includes a first metal portion and a second metal portion. The first metal portion forms a first junction with a first portion of the p-type semiconductor layer, and the second metal portion forms a second junction with a second portion of the p-type semiconductor layer. The energy barrier height of the first junction is different from the energy barrier height of the second junction. For example, the first junction may be a high-barrier Schottky (HBS) junction, and the second junction may be a low-barrier junction such as a low-barrier Schottky (LBS) junction or an ohmic junction. In some examples, the HBS junction has a barrier height greater than 1.7 eV, and the LBS junction has a barrier height lower than 1.7 eV.

[0095] The ohmic junction and/or low-barrier junction may provide a Kelvin connection to the p-type semiconductor layer to remove the floating, quasi-neutral node and reduce the voltage drop between the drain electrical contact and the floating, quasi-neutral node. Reducing the voltage drop may improve the hole injection efficiency of the p-type drain contact structure for mitigating current collapse and reducing dynamic on-state resistance (dR_{on}). Due to the improved hole injection efficiency of the p-type drain contact structure, the p-type drain contact structure can be smaller to reduce the depleted region of the channel layer under the p-type drain contact structure (and the reduced channel width penalty), thereby reducing the static on-state resistance $R_{\text{sub.DSON}}$. As such, the channel length does not need to be reduced in order to reduce the static on-state resistance $R_{\text{sub.DSON}}$. The smaller p-type drain contact structures may also simplify the layout. In addition, the voltage drop at the drain can be the same under DC conditions and in

switching events. Therefore, the dynamic on-state resistance may be closer to the static on-state resistance, as shown by a plot **430** in FIG. **4**.

[0096] FIG. **5A** is a cross-sectional view of an example of an HEMT **500** including a hybrid drain contact structure according to certain examples. In the illustrated example, HEMT **500** includes a substrate **505**, a channel layer **510** (e.g., including a GaN layer) grown on substrate **505**, a barrier layer **520** (e.g., including an AlGa_N layer) over channel layer **510**, and drain, source, and gate structures. The drain, source, and gate structures are isolated by the dielectric material of a dielectric layer **560**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. In some examples, HEMT **500** shown in FIG. **5A** may be a half pitch of an HEMT device that includes HEMT **500** and a mirrored version of HEMT **500** that shares the drain structure with HEMT **500**.

[0097] Substrate **505** may be a bulk semiconductor substrate, a semiconductor-on-insulator (SOI) substrate, or any other suitable substrate. In one example, substrate **505** may include a bulk silicon wafer, and may include one or more transition layers or buffer layers of suitable materials for accommodating the lattice mismatch between substrate **505** and channel layer **510** (e.g., to reduce or minimize lattice defect generation and/or propagation in channel layer **510**). For example, the transition layers or buffer layers may have a gradient concentration of one or more elements in a surface normal direction (e.g., z direction) of substrate **505** to gradually change the lattice constant.

[0098] Channel layer **510** and barrier layer **520** may be epitaxially grown on substrate **505** to form a heterostructure that may induce a 2DEG **512** near the interface between channel layer **510** and barrier layer **520** due to the different energy band structures of channel layer **510** and barrier layer **520**. 2DEG **512** may conduct current in a two-dimensional plane (e.g., an x-y plane). In some examples, channel layer **510** may be a portion of substrate **505**. In the illustrated example, channel layer **510** includes a GaN layer. In some examples, the material of channel layer **510** includes an unintentionally doped material, such as a material doped by diffusion of dopants from another layer, or includes an intrinsic material. In the illustrated example, barrier layer **520** includes an AlGa_N layer. Other materials may also be used for channel layer **510** and barrier layer **520**. For example, channel layer **510** may include indium aluminum gallium nitride (In.sub.iAl.sub.jGa.sub.1-i-jN) (where 0≤i≤1, 0≤j≤1, and 0≤i+j≤1), and barrier layer **520** may include indium aluminum gallium nitride (In.sub.kAl.sub.lGa.sub.1-k-lN) (where 0≤k≤1, 0≤l≤1, and 0≤k+l≤1).

[0099] The gate structure of HEMT **500** includes a gate semiconductor layer **530** over an upper surface of barrier layer **520**. In some examples, gate semiconductor layer **530** includes a p-doped semiconductor layer. For example, gate semiconductor layer **530** may include a GaN layer, or more generally, an In.sub.mAl.sub.nGa.sub.1-m-nN layer (where 0≤m<1, 0≤n<1, and 0≤m+n≤1). The p-type dopant with which gate semiconductor layer **530** is doped may include magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof. In examples where gate semiconductor layer **530** includes GaN doped with a p-type dopant, gate semiconductor layer **530** may be referred to as a p-GaN layer. Further, in examples where gate semiconductor layer **530** includes GaN doped with magnesium, gate semiconductor layer **530** may be referred to as a magnesium doped gallium nitride (Ga_N:Mg) layer. In some examples, a concentration of the dopant that is electrically activated in gate semiconductor layer **530** may be equal to or greater than about 1×10¹⁷ cm.⁻³. In some examples, the concentration may be equal to or greater than about 1×10¹⁸ cm.⁻³. Other materials, dopants, and/or concentrations may be used in other examples. Gate semiconductor layer **530** may be formed by epitaxial growth and selective etching using an etch mask, or may be formed by selective area growth using a growth mask. The etch mask or growth mask may define the shape and size of gate semiconductor layer **530**. The doping density and thickness of p-doped gate semiconductor layer **530** and the thickness of barrier layer **520** under gate semiconductor layer **530** may be selected such that the p-doped gate semiconductor layer **530** may deplete 2DEG **512** under gate semiconductor layer **530**, such that HEMT **500** is off

without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0100] A gate electrical contact **532** is formed on gate semiconductor layer **530** to transmit a gate voltage to gate semiconductor layer **530**. Gate electrical contact **532** may be electrically connected to a gate drive circuit through electrical interconnects such as conductive traces and/or vias (now shown). In the illustrated example, gate electrical contact **532** may laterally extend beyond gate semiconductor layer **530** to form a gate field plate **534**. Gate field plate **534** may be used in high voltage and low voltage GaN power devices to, for example, reduce current collapse and dynamic on-state resistance, and increase the breakdown voltage. Gate electrical contact **532** may include one or more metal and/or metal alloy materials having high electrical conductivity. Depending on the metal work function of gate electrical contact **532** and the energy band structure of gate semiconductor layer **530**, the metal-to-semiconductor contact between gate electrical contact **532** and gate semiconductor layer **530** may be, for example, an ohmic contact or a Schottky contact having a high or low barrier height. As described above, a Schottky contact between gate electrical contact **532** and gate semiconductor layer **530** may reduce gate leakage.

[0101] At the source region of HEMT **500**, a source electrical contact **540** extends through barrier layer **520** and contacts a source region of channel layer **510**. Source electrical contact **540** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **510**. One or more source field plates **542** and **544** may be formed in a dielectric layer **560** and may be coupled to source electrical contact **540**. The source field plates may be used to reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **500**.

[0102] At the drain region of HEMT **500**, a drain electrical contact **550** extends through dielectric layer **560** and barrier layer **520** and contacts a drain region of channel layer **510**. Drain electrical contact **550** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **510**. In addition to drain electrical contact **550**, the drain structure of HEMT **500** also includes a p-type drain contact structure that includes a drain semiconductor layer **552** over the upper surface of barrier layer **520**. Drain semiconductor layer **552** may be separate from drain electrical contact **550**, or may contact drain electrical contact **550**. In some examples, drain semiconductor layer **552** includes a p-doped semiconductor layer, such as a p-GaN layer. In some examples, drain semiconductor layer **552** and gate semiconductor layer **530** may be fabricated using the same processes. For example, a p-doped semiconductor layer may be epitaxially grown over barrier layer **520** and then selectively etched using an etch mask that defines the shapes and sizes of gate semiconductor layer **530** and drain semiconductor layer **552**. As such, gate semiconductor layer **530** and drain semiconductor layer **552** may include the same material and have the same thickness.

[0103] In some examples, an additional epitaxial growth process may be performed to further grow gate semiconductor layer **530** such that gate semiconductor layer **530** is thicker than drain semiconductor layer **552**. In some examples, barrier layer **520** under gate semiconductor layer **530** may be etched before the epitaxial growth of the p-doped semiconductor layer, such that barrier layer **520** under gate semiconductor layer **530** may be thinner than barrier layer **520** under drain semiconductor layer **552**, while gate semiconductor layer **530** may be thicker than drain semiconductor layer **552**. In this way, 2DEG **512** under gate semiconductor layer **530** may be completely depleted by the p-doped gate semiconductor layer **530** to turn off HEMT **500**, while 2DEG **512** under drain semiconductor layer **552** may be partially depleted and thus can still conduct current.

[0104] In the illustrated example, a drain metal contact is formed on drain semiconductor layer **552**. The drain metal contact may be electrically connected to drain electrical contact **550** and/or may be connected to a drain drive circuit through electrical interconnects. The drain metal contact includes a first metal portion **554** and a second metal portion **556**. Second metal portion **556** is

electrically connected to drain electrical contact 550 and may, in some examples, have the same material as drain electrical contact 550. First metal portion 554 is on a first semiconductor region of drain semiconductor layer 552. Second metal portion 556 is on a second semiconductor region of drain semiconductor layer 552. In some examples, first metal portion 554 may be much larger than second metal portion 556. First metal portion 554 and the first semiconductor region of drain semiconductor layer 552 form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce leakage current of the p-type drain contact structure during operation of HEMT 500. Second metal portion 556 and the second semiconductor region of drain semiconductor layer 552 form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction.

[0105] An energy barrier height of a junction between a metal and a semiconductor material can be a function of the materials of the metal and semiconductor material and any doping of the semiconductor material. For example, the energy barrier height of a metal-to-semiconductor junction can be determined by the metal work function of the metal and the energy band structure of the semiconductor material. In some examples, to achieve the different energy barrier heights, first metal portion 554 and second metal portion 556 may include different metal or metal alloy materials. In some examples, first metal portion 554 of the drain metal contact has a work function that is greater than a work function of second metal portion 556 of the drain metal contact. For example, the work function of first metal portion 554 of the drain metal contact is equal to or greater than about 4.6 electron volt (eV), whereas the work function of second metal portion 556 of the drain metal contact is less than about 4.6 eV. In some examples, the energy barrier height of the first junction is equal to or greater than about 1.7 eV, and the energy barrier height of the second junction is less than about 1.7 eV. In some examples, the first junction may be a high-barrier Schottky junction, and the second junction may be an ohmic junction (e.g., having an energy barrier height of about 0 eV) or may be a low-barrier Schottky junction. Within the drain metal contact, first metal portion 554 may form an ohmic junction with second metal portion 556, which may be contiguous with drain electrical contact 550 or may be electrically connected to drain electrical contact 550 by an electrical interconnect such as a trace or wire.

[0106] In the illustrated example, the semiconductor material and doping condition of the first semiconductor region of drain semiconductor layer 552 are the same as those of the second semiconductor region of drain semiconductor layer 552. For example, drain semiconductor layer 552 may be a magnesium doped gallium nitride (GaN:Mg) layer with magnesium (Mg) doped at an activated concentration about 2×10^{18} cm⁻³. First metal portion 554 may include, for example, titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (Ta₂N₅), copper (Cu), tungsten (W), or an alloy thereof. Second metal portion 556 may include, for example, gold (Au), aluminum (Al), an alloy, or a combination thereof. In some examples, the alloy may include titanium tungsten aluminum (TiWAl) or titanium aluminum nitride (TiAlN), each with an increased percentage of aluminum. In some examples, first metal portion 554 and second metal portion 556 include different metal materials, where the metal materials can include any of the metal materials described above, such as titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (Ta₂N₅), copper (Cu), tungsten (W), gold (Au), aluminum (Al), titanium tungsten aluminum (TiWAl), titanium aluminum nitride (TiAlN), or a combination thereof.

[0107] In some other examples, to achieve the different energy barrier heights, first metal portion 554 and second metal portion 556 can include the same metal or metal alloy material, or different metal materials having similar metal work functions, but the first semiconductor region and the second semiconductor region of drain semiconductor layer 552 may have different compositions, such as different p-type dopants, different doping densities, different activation levels of the dopant, or a combination thereof. Therefore, the first semiconductor region and the second semiconductor region of drain semiconductor layer 552 may have different energy band structures, and thus may

form junctions with different energy barrier heights with a same metal material. The metal material can include, for example, titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), titanium tungsten aluminum (TiWAl), titanium aluminum nitride (TiAlN), or a combination thereof. In some examples, first metal portion **554** and second metal portion **556** can be processed with different process conditions (e.g., different anneal conditions, different deposition/etch conditions, etc.) to create junctions of different barrier heights.

[0108] Dielectric layer **560** includes one or more dielectric materials that isolate and protect the gate structure, drain structure, and source structure. Dielectric layer **560** may include multiple dielectric layers of a same dielectric material or different dielectric materials deposited in one or more deposition processes. For example, dielectric layer **560** may include an oxide-based material or a nitride-based material, such as silicon oxide (e.g., a phosphosilicate glass (PSG)), aluminum oxide, silicon nitride, and the like. In some examples, dielectric layer **560** may further include one or more etch stop layers, such as silicon nitride (SiN) and the like, for controlling the etch depth of etching processes (e.g., for patterning a dielectric or metal layer).

[0109] In some examples, the electrical contacts or other metal electrical interconnects described above may each include one or more metal barrier layers and/or one or more adhesion layers (e.g., titanium nitride (TiN), tantalum nitride (TaN), and the like, or a combination thereof) between the metal material (e.g., aluminum (Al), copper (Cu), tungsten (W), the like, or a combination thereof) and dielectric layer **560**. The one or more metal barrier layers may prevent the diffusion of metal atoms into dielectric layer **560**. The one or more adhesion layers may be used to improve the adhesion of the metal material to the dielectric material of dielectric layer **560** to reduce or avoid defects and reliability issues such as interfacial delamination.

[0110] As described above and shown in FIG. 5A, the p-type drain contact structure including a first junction formed by a p-doped semiconductor layer and a metal contact can be used to inject holes to the barrier layer, the channel layer, and the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse. When the first junction includes a Schottky junction with a high barrier height (e.g., >1.7 eV), the leak current of the p-type drain contact structure may be low, but the hole injection efficiency of the p-type drain contact structure may also be low due to, for example, the dividing of the drain voltage by the back-to-back diodes and thus the reduction of the voltage level applied to the p-type semiconductor layer. Therefore, the normalized dynamic on-state resistance (dR_{on}) may still be significantly higher than 1 at high drain drive voltages. Adding the second junction with a lower energy barrier height, such as an ohmic junction, to the p-type drain contact structure can provide a Kelvin connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height, thereby increasing the voltage level at the p-type semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure. Due to the improved hole injection efficiency, the size of drain semiconductor layer **552** can be reduced and can still achieve the same hole injection and current collapse mitigation results. As such, the depleted region of the channel layer under the p-type drain contact structure can be reduced. Therefore, the channel width of the HEMT may be higher, and thus the static on-state resistance can be lower and the channel length does not need to be reduced to reduce the static $R_{sub,DS}$. When the channel length and thus the gate-drain spacing are longer, the breakdown voltage of the device can be higher.

[0111] FIG. 5B illustrates differences between an example of a hybrid drain contact structure including Schottky junctions and an example of a hybrid drain contact structure including two or more junctions with different barrier heights according to certain examples. The left diagram in FIG. 5B is a top view of a hybrid drain contact structure **502** that includes a plurality of ohmic drain contact structures **506** and a plurality of p-type drain contact structures **508** as described above with respect to, for example, FIG. 3B. Ohmic drain contact structures **506** may be in direct

contact with the channel layer as described above with respect to drain contact structure **304**. Each p-type drain contact structure **508** may include a p-GaN block and a metal layer that form a metal-to-semiconductor junction such as a Schottky junction, as described above with respect to, for example, FIGS. **3A-3E**. In the illustrated example, the p-GaN block of each p-type drain contact structure **508** has a hexagonal shape and tapers in a direction towards a gate structure. In other examples, the p-GaN block can have any other suitable regular, irregular, or arbitrary shapes. As described above, hybrid drain contact structure **502** shown in FIG. **3B** may be the common drain contact structure of an HEMT that includes two symmetrical half pitches sharing the common drain contact structure.

[0112] The right diagram in FIG. **5B** is a top view of an example of a hybrid drain contact structure **504** that includes a plurality of drain electrical contacts **550** and a plurality of p-type drain contact structures. Hybrid drain contact structure **504** may be an example of the hybrid drain contact structure shown in FIG. **5A**. As shown in FIGS. **5A** and **5B**, each p-type drain contact structure may include drain semiconductor layer **552**, and first metal portion **554** and second metal portion **556** formed on drain semiconductor layer **552**. As described above, first metal portion **554** and drain semiconductor layer **552** may form a first junction having a high barrier height (e.g., a Schottky junction), while second metal portion **556** and drain semiconductor layer **552** may form a second junction having a low barrier height, such as an ohmic contact or a low-barrier Schottky junction. In the illustrated example, drain semiconductor layer **552** has a hexagonal shape and tapers in a direction towards the gate structure. First metal portion **554** on drain semiconductor layer **552** may have a shape matching the shape of drain semiconductor layer **552**, and may also taper in a direction towards the gate structure. Second metal portions **556** and drain electrical contacts **550** may interleave and may be contiguous to form a continuous ohmic contact. Second metal portions **556** and drain electrical contacts **550** may have the same length or different lengths in the x direction. In some examples, second metal portions **556** and drain electrical contacts **550** may include a same metal material. Hybrid drain contact structure **504** shown in FIG. **5B** may be the common drain contact structure of an HEMT that includes two symmetrical half pitches sharing the common drain contact structure.

[0113] FIG. **5C** is a cross-sectional view of an example of a hybrid drain contact structure including a Schottky contact and a low-barrier (e.g., ohmic) contact as shown in FIG. **5A**, with an overlaid circuit model of the hybrid drain contact structure according to certain examples. The hybrid drain contact structure and the corresponding circuit model shown in FIG. **5C** may be similar to those of the hybrid drain contact structure of FIG. **3E**, but may include a low-barrier junction in addition to the higher barrier height junction (e.g., Schottky junction). As illustrated, the hybrid drain contact structure includes drain electrical contact **550** that forms an ohmic contact with channel layer **510**, first metal portion **554** that forms a Schottky contact with a first region of drain semiconductor layer **552**, and second metal portion **556** that forms a lower barrier height junction with a second region of drain semiconductor layer **552**.

[0114] In the circuit model, a drain contact node **562** is in first metal portion **554**, and a p-type drain node **576** is in drain semiconductor layer **552**. One or more channel nodes, such as a first channel node **584** and a second channel node **590**, are at an interface between channel layer **510** and barrier layer **520**. A Schottky diode **570** and a Schottky capacitor **572** are formed by first metal portion **554** and drain semiconductor layer **552**, and are electrically coupled between drain contact node **562** and p-type drain node **576**. The cathode terminal of Schottky diode **570** is electrically coupled to drain contact node **562**, and the anode terminal of Schottky diode **570** is electrically coupled to the p-type drain node **576**. Similarly, a first terminal of Schottky capacitor **572** is electrically coupled to drain contact node **562**, and a second terminal of Schottky capacitor **572** is electrically coupled to p-type drain node **576**.

[0115] A first side diode **580** (e.g., a p-i-n diode) and a first side capacitor **582** are formed by drain semiconductor layer **552**, barrier layer **520**, and channel layer **510**. First side diode **580** and first

side capacitor **582** are electrically coupled between p-type drain node **576** and first channel node **584**. The anode terminal of first side diode **580** is electrically coupled to p-type drain node **576**, and the cathode terminal of first side diode **580** is electrically coupled to first channel node **584**. A first terminal of first side capacitor **582** is electrically coupled to p-type drain node **576**, and a second terminal of first side capacitor **582** is electrically coupled to first channel node **584**.

[0116] In examples where drain semiconductor layer **552** is separate from drain electrical contact **550** that forms the ohmic contact with channel layer **510**, a second side diode **586** (e.g., a p-i-n diode) and a second side capacitor **588** may be formed by drain semiconductor layer **552**, barrier layer **520**, and channel layer **510**. Second side diode **586** and second side capacitor **588** may be electrically coupled between p-type drain node **576** and second channel node **590**. The anode terminal of second side diode **586** may be electrically coupled to p-type drain node **576**, and the cathode terminal of second side diode **586** may be electrically coupled to second channel node **590**. A first terminal of second side capacitor **588** may be electrically coupled to p-type drain node **576**, and a second terminal of second side capacitor **588** may be electrically coupled to second channel node **590**.

[0117] The circuit model of the hybrid drain contact structure may also include a field effect transistor (FET) **578** having a gate terminal electrically coupled to p-type drain node **576**. The source terminal of FET **578** may be electrically coupled to first channel node **584**, and the drain terminal of FET **578** may be electrically coupled to second channel node **590**.

[0118] As illustrated by the circuit model, the hybrid drain contact structure with a Schottky junction between first metal portion **554** and drain semiconductor layer **552** includes a back-to-back diode configuration at p-type drain node **576**, where the anode terminals of Schottky diode **570** and first side diode **580** (and second side diode **586**) are electrically coupled to a same node (e.g., p-type drain node **576**). Such a back-to-back diode configuration may otherwise result in a voltage divider and a floating, quasi-neutral node (e.g., p-type drain node **576**) in drain semiconductor layer **552** under some circumstances. For example, the floating, quasi-neutral node in drain semiconductor layer **552** may be in a voltage divider formed by Schottky diode **570** and first side diode **580** (and/or second side diode **586**). The voltage divider may divide the voltage level applied to first metal portion **554**, such that the effective voltage level applied to drain semiconductor layer **552** is lower than the voltage level applied to first metal portion **554**, and thus the voltage applied across the first side diode **580** (and/or second side diode **586**) is reduced. As a result, the drive strength and the hole injection efficiency of the hybrid drain contact structure may be low.

[0119] As shown in FIG. 5C, the circuit model of the hybrid drain contact structure described in FIG. 5A includes an ohmic resistor **574** electrically coupled between second metal portion **556** and drain semiconductor layer **552** to model the ohmic junction between second metal portion **556** and drain semiconductor layer **552**. Ohmic resistor **574** may provide a Kelvin connection to p-type drain node **576** (e.g., drain semiconductor layer **552**), to reduce the voltage drop (e.g., to about 0 V) between second metal portion **556** and p-type drain node **576**, and remove the floating, quasi-neutral node. Reducing the voltage drop may effectively increase the voltage level at drain semiconductor layer **552**, thereby improving the hole injection efficiency of the p-type drain contact structure for mitigating current collapse and reducing dynamic on-state resistance. For example, as described above, when a positive voltage level is applied to drain electrical contact **550** and the source is at a low voltage level, the voltage level in the channel layer may gradually decrease from the drain region to the gate and source regions. When the voltage level applied to drain electrical contact **550** is sufficiently high, a voltage difference between the voltage level applied to drain electrical contact **550** (and thus the voltage level at drain semiconductor layer **552** due to the very small voltage drop on ohmic resistor **574**) and the voltage level at the region of the channel layer below the p-type drain contact structure, which is the effective voltage level applied across first side diode **580** (or second side diode **586**), may be sufficiently high to turn on first side

diode **580** (and second side diode **586**) to inject holes to barrier layer **520** and channel layer **510**.

[0120] Due to the improved hole injection efficiency of the p-type drain contact structure, the p-type drain contact structure can be smaller to reduce the depleted region of the channel layer under the p-type drain contact structure, thereby reducing the static on-state resistance. As such, the channel length does not need to be reduced to reduce the static on-state resistance, and thus the breakdown voltage of the HEMT can be high. The smaller p-type drain contact structures may also simplify the layout. In addition, the voltage drop can be the same under DC conditions and in switching events. Therefore, the dynamic on-state resistance may be closer to the static on-state resistance, as shown by a curve **430** in FIG. **4**. In addition, Schottky capacitor **572** may be bypassed by ohmic resistor **574**, and thus the drain current may be more stable in switching events.

[0121] FIG. **6A** illustrates an example of an enhancement mode HEMT **600** including a hybrid drain contact structure having a Schottky contact and an ohmic contact according to certain examples. HEMT **600** may be an example of HEMT **500**. In the illustrated example, HEMT **600** includes a substrate (not shown), a channel layer **610** (e.g., including a GaN layer), a barrier layer **620** (e.g., including an AlGaIn) layer, and drain, source, and gate structures. The drain, source, and gate structures are isolated and protected by the dielectric material of a dielectric layer **660**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. In some examples, HEMT **600** shown in FIG. **6A** may be a half pitch of an HEMT device that includes HEMT **600** and a mirrored version of HEMT **600** that shares the drain structure with HEMT **600**.

[0122] The substrate of HEMT **600** may be similar to substrate **505** described above, and thus is not described in detail with respect to FIG. **6A**. Channel layer **610** and barrier layer **620** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **612** near the interface between channel layer **610** and barrier layer **620** due to the different energy band structures of channel layer **610** and barrier layer **620**. Channel layer **610**, barrier layer **620**, and 2DEG **612** may be similar to channel layer **510**, barrier layer **520**, and 2DEG **512**, respectively. In the illustrated example, channel layer **610** includes a GaN layer, and barrier layer **620** includes an AlGaIn layer. Other materials may also be used for channel layer **610** and barrier layer **620** as described above with respect to FIG. **5A**.

[0123] The gate structure of HEMT **600** includes a gate semiconductor layer **630** over an upper surface of barrier layer **620**. Gate semiconductor layer **630** may be similar to gate semiconductor layer **530** and includes a p-doped semiconductor layer, such as a gallium nitride layer doped with magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof, as described above with respect to FIG. **5A**. Gate semiconductor layer **630** may be formed by epitaxial growth and selective etching using an etch mask, or may be formed by selective area growth using a growth mask. The etch mask or growth mask may define the shape and size of gate semiconductor layer **630**. The doping density and thickness of p-doped gate semiconductor layer **630** and the thickness of barrier layer **620** under gate semiconductor layer **630** may be selected such that the p-doped gate semiconductor layer **630** may deplete 2DEG **612** under gate semiconductor layer **630**, such that HEMT **600** is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0124] A first gate electrical contact **632** is deposited on a first semiconductor portion of gate semiconductor layer **630** to function as a gate electrode for applying a gate voltage to gate semiconductor layer **630**. First gate electrical contact **632** may be electrically connected to a gate drive circuit (now shown). First gate electrical contact **632** and the first semiconductor portion of gate semiconductor layer **630** may form a Schottky contact, which may reduce gate leakage. A second gate electrical contact **634** is deposited on a second semiconductor portion of gate semiconductor layer **630**. Second gate electrical contact **634** and the second semiconductor portion of gate semiconductor layer **630** may form an ohmic contact, which may reduce the voltage drop between the first and second gate electrical contacts **632** and **634** and gate semiconductor layer **630**.

under DC conditions and in switching events, thereby reducing the effective gate overdrive voltage (V_{sub}.GT). In the illustrated example, the gate structure also includes a gate field plate **636**. Gate field plate **636** may be used in high voltage and low voltage GaN power devices to, for example, reduce current collapse and dynamic on-state resistance, and increase the breakdown voltage of HEMT **600**. First gate electrical contact **632**, second gate electrical contact **634**, and gate field plate **636** may include one or more metal and/or metal alloy materials having high electrical conductivity.

[0125] At the source region of HEMT **600**, a source electrical contact **640** extends through barrier layer **620** and contacts a source region of channel layer **610**. Source electrical contact **640** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **610**. One or more source field plates **642** and **644** may be formed in dielectric layer **660** and may be coupled to source electrical contact **640**. As described above, the field plates may be used to reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **600**.

[0126] At the drain region of HEMT **600**, a drain electrical contact **650** extends through dielectric layer **660** and barrier layer **620**, and contacts a drain region of channel layer **610**. Drain electrical contact **650** may include a metal or metal alloy, and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **610**. In addition to drain electrical contact **650**, the drain structure of HEMT **600** also includes a p-type drain contact structure that includes a drain semiconductor layer **652** over the upper surface of barrier layer **620**. Drain semiconductor layer **652** may be separate from drain electrical contact **650**, or may contact drain electrical contact **650**. In some examples, drain semiconductor layer **652** includes a p-doped semiconductor layer, such as a p-GaN layer. In the illustrated example, drain semiconductor layer **652** and gate semiconductor layer **630** may be fabricated using the same processes. For example, a p-doped semiconductor layer may be epitaxially grown on barrier layer **620** and then selectively etched using an etch mask that defines the shapes and sizes of gate semiconductor layer **630** and drain semiconductor layer **652**. As such, gate semiconductor layer **630** and drain semiconductor layer **652** may include the same material and have the same thickness.

[0127] In the illustrated example, a drain metal contact is deposited on drain semiconductor layer **652**. The drain metal contact may be electrically connected to drain electrical contact **650** and/or may be connected to a drain drive circuit. The drain metal contact includes a first metal portion **654** and a second metal portion **656**. Second metal portion **656** is electrically connected to drain electrical contact **650** and may, in some examples, contact drain electrical contact **650** and/or have the same material as drain electrical contact **650**. First metal portion **654** is on a first semiconductor region of drain semiconductor layer **652**. Second metal portion **656** is on a second semiconductor region of drain semiconductor layer **652**. In some examples, first metal portion **654** may be larger than second metal portion **656**. First metal portion **654** and the first semiconductor region of drain semiconductor layer **652** may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure and provide some current collapse mitigation capability. Second metal portion **656** and the second semiconductor region of drain semiconductor layer **652** may form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction, to improve the hole injection efficiency of the -type drain contact structure. First metal portion **654** and second metal portion **656** may include different metal or metal alloy materials that have different work functions. First metal portion **654** may include, for example, titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (Ta₂N₃), copper (Cu), tungsten (W), or an alloy thereof. Second metal portion **656** may include, for example, gold (Au), aluminum (Al), or an alloy thereof. In some examples, the alloy may include titanium tungsten aluminum (TiWAl) or titanium aluminum nitride (TiAlN). In some examples, first metal portion **654** and second metal portion **656** may include any of the metal

materials described above, such as titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), titanium tungsten aluminum (TiWAl), titanium aluminum nitride (TiAlN), or a combination thereof.

[0128] Dielectric layer **660** includes dielectric materials that isolate and protect the gate structure, drain structure, and source structure. Dielectric layer **660** may include multiple dielectric layers of a same dielectric material or different dielectric materials deposited in one or more deposition processes. For example, dielectric layer **660** may include an oxide-based material or a nitride-based material, such as silicon oxide (e.g., a phosphosilicate glass (PSG)), aluminum oxide, silicon nitride, and the like. In some examples, dielectric layer **660** may further include one or more etch stop layers, such as silicon nitride (SiN) and the like, for controlling the etch depth of etching processes (e.g., for patterning a dielectric or metal layer). In some examples, the electrical contacts or other metal electrical interconnects described above may each include one or more metal barrier layers and/or one or more adhesion layers (e.g., titanium nitride (TiN), tantalum nitride (TaN), and the like, or a combination thereof) between the metal material (e.g., aluminum (Al), copper (Cu), tungsten (W), the like, or a combination thereof) and dielectric layer **660**.

[0129] As described above, the p-type drain contact structure including a first junction formed by a p-type semiconductor layer and a metal contact may be used to inject holes to the barrier layer, the channel layer, and the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse. When the first junction includes a Schottky junction with a high barrier height (e.g., >1.7 eV), the leak current of the p-type drain contact structure may be reduced, but the hole injection efficiency of the p-type drain contact structure may be low due to, for example, the dividing of the drain voltage by the back-to-back diodes and thus the reduction of the voltage level applied to the p-type semiconductor layer. Therefore, the normalized dynamic on-state resistance (dRon) may still be significantly higher than 1 at high drain drive voltages as described above with respect to FIGS. 4-5C. Adding the second junction with a lower energy barrier height, such as an ohmic junction, to the p-type drain contact structure can provide a Kelvin connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height (e.g., the Schottky junction), thereby increasing the voltage level at the p-type semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure. Due to the improved hole injection efficiency, the size of drain semiconductor layer **652** can be reduced and can still achieve the same hole injection and current collapse mitigation results. As such, the depleted region of the channel layer under the p-type drain contact structure can be reduced. Therefore, the channel width of the HEMT may be higher, and thus the static on-state resistance can be lower and the channel length does not need to be reduced to reduce the static on-state resistance. When the channel length and thus the gate-drain spacing are longer, the breakdown voltage of the device can be higher.

[0130] The hybrid drain structures described with respect to FIGS. 5A and 6A may also be used in other enhancement mode transistor devices that are not p-GaN based enhancement mode HEMTs. For example, the hybrid drain structures disclosed herein may also be used in gate-injection transistor (GIT) enhancement mode devices, etched gate-barrier based enhancement mode devices, fluorine-implanted enhancement mode devices, or enhancement mode devices implemented using other techniques.

[0131] FIG. 6B illustrates an example of a depletion mode HEMT **602** including a hybrid drain contact structure having a Schottky contact and an ohmic contact according to certain examples. As HEMT **600**, HEMT **602** includes a substrate (not shown), channel layer **610** (e.g., including a GaN layer), barrier layer **620** (e.g., including an AlGaN) layer, and drain, source, and gate structures. Channel layer **610** and barrier layer **620** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **614** near the interface between channel layer **610** and

barrier layer **620** due to the different energy band structures of channel layer **610** and barrier layer **620**. The drain, source, and gate structures are isolated and protected by the dielectric material of dielectric layer **660**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. HEMT **602** has a gate structure different from the gate structure of HEMT **600**, and is an enhancement mode HEMT because of the gate structure that does not include a p-doped semiconductor layer to deplete the 2DEG channel under the gate. The source structure and the drain structure of HEMT **602** may be similar to the source structure and the drain structure of HEMT **602**, respectively. In some examples, HEMT **602** shown in FIG. **6B** may be a half pitch of an HEMT device that includes HEMT **602** and a mirrored version of HEMT **602** that shares the drain structure with HEMT **602**.

[0132] As illustrated, the gate structure of HEMT **602** includes a gate metal layer **638** separated from barrier layer **620** by a dielectric layer. When no negative voltage signal is applied to gate metal layer **638**, 2DEG **614** is not depleted and can conduct current between the drain and source, and HEMT **602** is turned on. When a negative voltage level is applied to gate metal layer **638**, electrons in the region of 2DEG **614** under gate metal layer **638** may be depleted to turn off HEMT **602**.

[0133] At the source region of HEMT **602**, a source electrical contact **640** extends through barrier layer **620** and contacts a source region of channel layer **610**. Source electrical contact **640** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **610**. One or more source field plates **642** and **644** may be formed in dielectric layer **660** and may be coupled to source electrical contact **640**. As described above, the field plates may reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **602**.

[0134] The drain structure of HEMT **602** includes drain electrical contact **650** that extends through dielectric layer **660** and barrier layer **620** and contacts the drain region of channel layer **610**. In addition to drain electrical contact **650**, the drain structure of HEMT **602** also includes a p-type drain contact structure that includes drain semiconductor layer **652** (e.g., a p-doped GaN layer) over the upper surface of barrier layer **620**. Drain semiconductor layer **652** may be separate from drain electrical contact **650**, or may contact drain electrical contact **650**. Drain semiconductor layer **652** may be fabricated by, for example, epitaxially growing a p-doped semiconductor layer over barrier layer **620** and selectively etching the p-doped semiconductor layer using an etch mask that defines the shape and size of drain semiconductor layer **652**. A drain metal contact is deposited on drain semiconductor layer **652**. The drain metal contact may be electrically connected to drain electrical contact **650** and/or may be connected to a drain drive circuit. The drain metal contact includes first metal portion **654** and second metal portion **656**. Second metal portion **656** is electrically connected to drain electrical contact **650** and may, in some examples, have the same material as drain electrical contact **650**. First metal portion **654** is on a first semiconductor region of drain semiconductor layer **652**. Second metal portion **656** is on a second semiconductor region of drain semiconductor layer **652**. First metal portion **654** and the first semiconductor region of drain semiconductor layer **652** forms a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure, and may have a limited capability for current collapse mitigation. Second metal portion **656** and the second semiconductor region of drain semiconductor layer **652** may form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction. First metal portion **654** and second metal portion **656** may include different metal or metal alloy materials that have different work functions as describe above.

[0135] The p-type drain contact structure including a first junction formed by a p-type semiconductor layer and a metal contact may be used to inject holes to the barrier layer, the channel layer, and/or the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse at high drain

voltages. When the first junction includes a Schottky junction with a high barrier height (e.g., >1.7 eV), the hole injection efficiency of the p-type drain contact structure may be low due to, for example, the dividing of the drain voltage by the back-to-back diodes and thus the reduction of the voltage level applied to the p-type semiconductor layer as described above. Adding the second junction with a lower energy barrier height, such as an ohmic junction, to the p-type drain contact structure can provide a Kelvin connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction (e.g., the Schottky diode), thereby increasing the voltage level at the p-type semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure. Due to the improved hole injection efficiency, the dynamic on-state resistance may be reduced, and the size of drain semiconductor layer **652** can be reduced and can still achieve the same hole injection and current collapse mitigation results. As such, the depleted region of the channel layer under the p-type drain contact structure can be reduced. Therefore, the channel width of the HEMT may be higher, and thus the static on-state resistance can be lower and the channel length does not need to be reduced to reduce the static on-state resistance. When the channel length and thus the gate-drain spacing are longer, the breakdown voltage of the device can be higher.

[0136] FIG. 7A illustrates an example of an enhancement mode HEMT **700** including a hybrid drain contact structure having a Schottky contact and a low-barrier Schottky (LBS) contact according to certain examples. HEMT **700** may be similar to HEMT **500** and HEMT **600**. In the illustrated example, HEMT **700** includes a substrate (not shown), a channel layer **710** (e.g., including a GaN layer), a barrier layer **720** (e.g., including an AlGaIn) layer, and drain, source, and gate structures. The drain, source, and gate structures are isolated and protected by the dielectric material of a dielectric layer **760**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. In some examples, HEMT **700** shown in FIG. 7A may be a half pitch of an HEMT device that includes HEMT **700** and a mirrored version of HEMT **700** that shares the drain structure with HEMT **700**.

[0137] The substrate of HEMT **700** may be similar to substrate **505** described above, and thus is not described in detail with respect to FIG. 7A. Channel layer **710** and barrier layer **720** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **712** near the interface between channel layer **710** and barrier layer **720** due to the different energy band structures of channel layer **710** and barrier layer **720**. Channel layer **710**, barrier layer **720**, and 2DEG **712** may be similar to channel layer **510**, barrier layer **520**, and 2DEG **512**, respectively. In the illustrated example, channel layer **710** includes a gallium nitride (GaN) layer, and barrier layer **720** includes an aluminum gallium nitride (AlGaIn) layer. Other materials may also be used for channel layer **710** and barrier layer **720** as described above with respect to FIG. 5A.

[0138] The gate structure of HEMT **700** includes a gate semiconductor layer **730** over an upper surface of barrier layer **720**. Gate semiconductor layer **730** may be similar to gate semiconductor layer **530** and includes a p-doped semiconductor layer, such as a GaN layer doped with magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof, as described above with respect to FIG. 5A. Gate semiconductor layer **730** may be formed by epitaxial growth and selective etching using an etch mask, or may be formed by selective area growth using a growth mask. The doping density and thickness of p-doped gate semiconductor layer **730** and the thickness of barrier layer **720** under gate semiconductor layer **730** may be selected such that the p-doped gate semiconductor layer **730** may deplete 2DEG **712** under gate semiconductor layer **730**, such that HEMT **700** is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0139] A first gate electrical contact **732** is deposited on a first semiconductor portion of gate semiconductor layer **730** to function as a gate electrode for applying a gate voltage to gate semiconductor layer **730**. First gate electrical contact **732** and the first semiconductor portion of gate semiconductor layer **730** may form a Schottky contact, which may reduce gate leakage. A

second gate electrical contact **734** is deposited on a second semiconductor portion of gate semiconductor layer **730**. Second gate electrical contact **734** and the second semiconductor portion of gate semiconductor layer **730** may form a low-barrier Schottky contact, which may reduce the voltage drop between the first and second gate electrical contacts **732** and **734** and gate semiconductor layer **730** under DC conditions and in switching events, thereby reducing the effective gate overdrive voltage ($V_{sub.GT}$). In the illustrated example, the gate structure also includes a gate field plate **736**. Gate field plate **736** may be used to, for example, reduce current collapse and dynamic on-state resistance, and increase the breakdown voltage of HEMT **700**. First gate electrical contact **732**, second gate electrical contact **734**, and gate field plate **736** may include one or more metal and/or metal alloy materials having high electrical conductivity.

[0140] At the source region of HEMT **700**, a source electrical contact **740** extends through barrier layer **720** and contacts a source region of channel layer **710**. Source electrical contact **740** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **710**. One or more source field plates **742** and **744** may be formed in dielectric layer **760** and may be coupled to source electrical contact **740**. As described above, the field plates may be used to, for example, reduce current collapse and dynamic on-state resistance, and/or increase the breakdown voltage of HEMT **700**.

[0141] At the drain region of HEMT **700**, a drain electrical contact **750** extends through dielectric layer **760** and barrier layer **720** and contacts a drain region of channel layer **710**. Drain electrical contact **750** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **710**. In addition to drain electrical contact **750**, the drain structure of HEMT **700** also includes a p-type drain contact structure that includes a drain semiconductor layer **752** (e.g., a p-GaN layer) over the upper surface of barrier layer **720**. Drain semiconductor layer **752** may be separate from drain electrical contact **750**, or may contact drain electrical contact **750**. In some examples, drain semiconductor layer **752** includes a p-doped semiconductor layer, such as a p-doped GaN layer. In the illustrated example, drain semiconductor layer **752** and gate semiconductor layer **730** may be fabricated using the same processes. For example, a p-doped semiconductor layer may be epitaxially grown on barrier layer **720** and then selectively etched using an etch mask that defines the shapes and sizes of gate semiconductor layer **730** and drain semiconductor layer **752**. As such, gate semiconductor layer **730** and drain semiconductor layer **752** may include the same material and have the same thickness.

[0142] A drain metal contact is deposited on drain semiconductor layer **752**. The drain metal contact may be electrically connected to drain electrical contact **750**. The drain metal contact includes a first metal portion **754** and a second metal portion **756**. Second metal portion **756** is electrically connected to drain electrical contact **750** and may, in some examples, have the same material as drain electrical contact **750** and contact drain electrical contact **750**. First metal portion **754** is on a first semiconductor region of drain semiconductor layer **752**. Second metal portion **756** is on a second semiconductor region of drain semiconductor layer **752**. In some examples, first metal portion **754** may be larger than second metal portion **756**. First metal portion **754** and the first semiconductor region of drain semiconductor layer **752** may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure and may have a limited capability for hole injection and current collapse mitigation. Second metal portion **756** and the second semiconductor region of drain semiconductor layer **752** may form a second junction that has a lower barrier height, such as a low-barrier Schottky junction. First metal portion **754** and second metal portion **756** may include different metal or metal alloy materials that have different work functions, and may include materials described above with respect to FIG. 5A.

[0143] Dielectric layer **760** includes dielectric materials that isolate and protect the gate structure, drain structure, and source structure. Dielectric layer **760** may include multiple dielectric layers of a same dielectric material or different dielectric materials deposited in one or more deposition

processes. For example, dielectric layer **760** may include an oxide-based material or a nitride-based material, such as silicon oxide (e.g., a phosphosilicate glass (PSG)), aluminum oxide, silicon nitride, and the like. In some examples, dielectric layer **760** may further include one or more etch stop layers, such as silicon nitride (SiN) and the like, for controlling the etch depth of etching processes (e.g., for patterning a dielectric or metal layer). In some examples, the electrical contacts or other metal electrical interconnects described above may each include one or more metal barrier layers and/or one or more adhesion layers (e.g., titanium nitride (TiN), tantalum nitride (TaN), and the like, or a combination thereof) between the metal material (e.g., aluminum (Al), copper (Cu), tungsten (W), the like, or a combination thereof) and dielectric layer **760**.

[0144] As described above, the p-type drain contact structure including a first junction formed by a p-type semiconductor layer and a metal contact may be used to inject holes to the barrier layer, the channel layer, and the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse. Adding the second junction with a lower energy barrier height, such as a low-barrier Schottky junction, to the p-type drain contact structure can provide a Kelvin connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height, thereby increasing the voltage level at the p-type semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure.

[0145] FIG. 7B illustrates an example of a depletion mode HEMT **702** including a hybrid drain contact structure having a Schottky contact and a low-barrier Schottky contact according to certain examples. HEMT **702** may be similar to HEMT **602** or **700** described above. As HEMT **700**, HEMT **702** includes a substrate (not shown), channel layer **710** (e.g., including a GaN layer), barrier layer **720** (e.g., including an AlGaIn) layer, and drain, source, and gate structures. Channel layer **710** and barrier layer **720** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **714** near the interface between channel layer **710** and barrier layer **720** due to the different energy band structures of channel layer **710** and barrier layer **720**. The drain, source, and gate structures are isolated by the dielectric material of dielectric layer **760**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. HEMT **702** has a gate structure different from the gate structure of HEMT **700**, and is an enhancement mode HEMT because of the different gate structure that does not include a p-doped semiconductor layer to deplete the 2DEG channel under the gate. The source structure and the drain structure of HEMT **702** may be similar to the source structure and the drain structure of HEMT **700**, respectively. In some examples, HEMT **702** shown in FIG. 7B may be a half pitch of an HEMT device that includes HEMT **702** and a mirrored version of HEMT **702** that shares the drain structure with HEMT **702**.

[0146] As illustrated, the gate structure of HEMT **702** includes a gate metal layer **738** separated from barrier layer **720** by a dielectric layer. When no negative voltage signal is applied to gate metal layer **738**, 2DEG **714** is not depleted and can conduct current between the drain and source, and HEMT **702** is turned on. When a negative voltage level is applied to gate metal layer **738**, electrons in the region of 2DEG **714** under gate metal layer **738** may be depleted to turn off HEMT **702**.

[0147] At the source region of HEMT **702**, a source electrical contact **740** extends through barrier layer **720** and contacts a source region of channel layer **710**. Source electrical contact **740** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **710**. One or more source field plates **742** and **744** may be formed in dielectric layer **760** and may be coupled to source electrical contact **740**. The field plates may be used to reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **702**.

[0148] The drain structure of HEMT **702** includes drain electrical contact **750** that extends through dielectric layer **760** and barrier layer **720** and contacts the drain region of channel layer **710**. In

addition to drain electrical contact **750**, the drain structure of HEMT **702** also includes a p-type drain contact structure that includes drain semiconductor layer **752** (e.g., a p-doped GaN layer) over the upper surface of barrier layer **720**. Drain semiconductor layer **752** may be separate from drain electrical contact **750**, or may contact drain electrical contact **750**. Drain semiconductor layer **752** may be fabricated by, for example, epitaxially growing a p-doped semiconductor layer over barrier layer **720** and selectively etching the p-doped semiconductor layer using an etch mask that defines the shape and size of drain semiconductor layer **752**. A drain metal contact is deposited on drain semiconductor layer **752**. The drain metal contact may be electrically connected to drain electrical contact **750** and a drain drive circuit. The drain metal contact includes first metal portion **754** and second metal portion **756**. Second metal portion **756** is electrically connected to drain electrical contact **750** and may, in some examples, have the same material as drain electrical contact **750**. First metal portion **754** is on a first semiconductor region of drain semiconductor layer **752**. Second metal portion **756** is on a second semiconductor region of drain semiconductor layer **752**.

[0149] First metal portion **754** and the first semiconductor region of drain semiconductor layer **752** forms a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure and may have some hole injection and current collapse mitigation capability as described above. Second metal portion **756** and the second semiconductor region of drain semiconductor layer **752** may form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction. First metal portion **754** and second metal portion **756** may include different metal or metal alloy materials that have different work functions as describe above. The p-type drain contact structure including the first junction formed by a p-doped semiconductor layer and a metal contact may be used to inject holes to the barrier layer, the channel layer, and/or the buffer layer at high drain voltages to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse at high drain voltages. Adding the second junction with a lower energy barrier height, such as an ohmic junction, to the p-type drain contact structure can provide a Kelvin connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height, thereby increasing the voltage level at the p-type semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure.

[0150] FIG. **8A** illustrates an example of an enhancement mode HEMT **800** including a hybrid drain contact structure having a Schottky contact, a low-barrier Schottky contact, and an ohmic contact according to certain examples. HEMT **800** may be similar to HEMTs **500**, **600**, and **700** described above. HEMT **800** includes a substrate (not shown), a channel layer **810** (e.g., including a GaN layer), a barrier layer **820** (e.g., including an AlGaIn layer), and drain, source, and gate structures. The drain, source, and gate structures are isolated and protected by the dielectric material of a dielectric layer **860**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. In some examples, HEMT **800** shown in FIG. **8A** may be a half pitch of an HEMT device that includes HEMT **800** and a mirrored version of HEMT **800** that shares the drain structure with HEMT **800**.

[0151] The substrate of HEMT **800** may be similar to substrate **505** described above, and thus is not described in detail with respect to FIG. **8A**. Channel layer **810** and barrier layer **820** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **812** near the interface between channel layer **810** and barrier layer **820** due to the different energy band structures of channel layer **810** and barrier layer **820**. Channel layer **810**, barrier layer **820**, and 2DEG **812** may be similar to channel layer **510**, barrier layer **520**, and 2DEG **512**, respectively. In the illustrated example, channel layer **810** includes a gallium nitride (GaN) layer, and barrier layer **820** includes an aluminum gallium nitride (AlGaIn) layer. Other materials may also be used for channel layer **810** and barrier layer **820** as described above with respect to FIG. **5A**.

[0152] The gate structure of HEMT **800** includes a gate semiconductor layer **830** over an upper surface of barrier layer **820**. Gate semiconductor layer **830** may be similar to gate semiconductor layer **530** and includes a p-doped semiconductor layer, such as a gallium nitride (GaN) layer doped with magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof, as described above with respect to FIG. 5A. Gate semiconductor layer **830** may be formed by epitaxial growth and selective etching using an etch mask, or may be formed by selective area growth using a growth mask. The doping density and thickness of p-doped gate semiconductor layer **830** and the thickness of barrier layer **820** under gate semiconductor layer **830** may be selected such that the p-doped gate semiconductor layer **830** may deplete 2DEG **812** under gate semiconductor layer **830**, such that HEMT **800** is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0153] A first gate electrical contact **832** is deposited on a first semiconductor portion of gate semiconductor layer **830** to function as a gate electrode for applying a gate voltage to gate semiconductor layer **830**. First gate electrical contact **832** and the first semiconductor portion of gate semiconductor layer **830** may form a Schottky contact, which may reduce gate leakage and provide some hole injection and current collapse mitigation capability at high drain voltages. A second gate electrical contact **834** is deposited on a second semiconductor portion of gate semiconductor layer **830**. Second gate electrical contact **834** and the second semiconductor portion of gate semiconductor layer **830** may form a low-barrier Schottky contact, which may reduce the voltage drop between the first and second gate electrical contacts **832** and **834** and gate semiconductor layer **830** under DC conditions and in switching events, thereby reducing the effective gate overdrive voltage ($V_{sub,GT}$). In the illustrated example, the gate structure also includes a gate field plate **836**. Gate field plate **836** may be used to, for example, reduce current collapse and dynamic on-state resistance, and increase the breakdown voltage of HEMT **800**. First gate electrical contact **832**, second gate electrical contact **834**, and gate field plate **836** may include one or more metal and/or metal alloy materials having high electrical conductivity.

[0154] At the source region of HEMT **800**, a source electrical contact **840** extends through barrier layer **820** and contacts a source region of channel layer **810**. Source electrical contact **840** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **810**. One or more source field plates **842** and **844** may be formed in dielectric layer **860** and may be coupled to source electrical contact **840**. As described above, the field plates may reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **800**.

[0155] At the drain region of HEMT **800**, a drain electrical contact **850** extends through dielectric layer **860** and barrier layer **820** and contacts a drain region of channel layer **810**. Drain electrical contact **850** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **810**. In addition to drain electrical contact **850**, the drain structure of HEMT **800** also includes a p-type drain contact structure that includes a drain semiconductor layer **852** (e.g., a p-doped GaN layer) over the upper surface of barrier layer **820**. Drain semiconductor layer **852** may be separate from drain electrical contact **850**, or may contact drain electrical contact **850**. In the illustrated example, drain semiconductor layer **852** and gate semiconductor layer **830** may be fabricated using the same processes as described with respect to, for example, FIGS. 5A, 6A, and 7A.

[0156] A drain metal contact is deposited on drain semiconductor layer **852**. The drain metal contact may be electrically connected to drain electrical contact **850**. The drain metal contact includes a first metal portion **854**, a second metal portion **856**, and a third metal portion **858**. Second metal portion **856** is electrically connected to drain electrical contact **850** and may, in some examples, have the same material as drain electrical contact **850** and contact drain electrical contact **850**. First metal portion **854** is on a first semiconductor region of drain semiconductor layer **852**. Second metal portion **856** is on a second semiconductor region of drain semiconductor layer **852**.

Third metal portion **858** is on a third semiconductor region of drain semiconductor layer **852**. First metal portion **854** and the first semiconductor region of drain semiconductor layer **852** may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure, and may be used to inject holes to the barrier layer, the channel layer, and the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse. Second metal portion **856** and the second semiconductor region of drain semiconductor layer **852** may form a second junction that has a lower barrier height, such as an ohmic junction. Third metal portion **858** and the third semiconductor region of drain semiconductor layer **852** may form a third junction that has a barrier height between the first junction and the second junction, such as a low-barrier Schottky junction. The second junction and the third junction can provide a low-resistance connection to the p-type semiconductor layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction, thereby increasing the voltage level at the p-GaN layer and thus the hole injection efficiency of the p-type drain contact structure. First metal portion **854**, second metal portion **856**, and third metal portion **858** may include different metal or metal alloy materials that have different work functions to achieve desired barrier heights, and may include materials described above with respect to FIG. 5A.

[0157] FIG. 8B illustrates an example of a depletion mode HEMT including a hybrid drain contact structure having a Schottky contact, a low-barrier Schottky contact, and an ohmic contact according to certain examples. HEMT **802** may be similar to HEMT **602** or **702** described above. As HEMT **800**, HEMT **802** includes a substrate (not shown), channel layer **810** (e.g., including a GaN layer), barrier layer **820** (e.g., including an AlGaN layer), and drain, source, and gate structures. Channel layer **810** and barrier layer **820** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **814** near the interface between channel layer **810** and barrier layer **820** due to the different energy band structures of channel layer **810** and barrier layer **820**. The drain, source, and gate structures are isolated by the dielectric material of dielectric layer **860**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. HEMT **802** has a gate structure different from the gate structure of HEMT **800**, and is an enhancement mode HEMT because of the different gate structure that does not include a p-doped semiconductor layer to deplete the 2DEG channel under the gate. The source structure and the drain structure of HEMT **802** may be similar to the source structure and the drain structure of HEMT **800**, respectively. In some examples, HEMT **802** shown in FIG. 8B may be a half pitch of an HEMT device that includes HEMT **802** and a mirrored version of HEMT **802** that shares the drain structure with HEMT **802**.

[0158] As illustrated, the gate structure of HEMT **802** includes a gate metal layer **838** separated from barrier layer **820** by a dielectric layer. When no negative voltage signal is applied to gate metal layer **838**, 2DEG **814** is not depleted and can conduct current between the drain and source, and HEMT **802** is turned on. When a negative voltage level is applied to gate metal layer **838**, electrons in the region of 2DEG **814** under gate metal layer **838** may be depleted to turn off HEMT **802**.

[0159] At the source region of HEMT **802**, a source electrical contact **840** extends through barrier layer **820** and contacts a source region of channel layer **810**. Source electrical contact **840** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **810**. One or more source field plates **842** and **844** may be formed in dielectric layer **860** and may be coupled to source electrical contact **840**. As described above, the field plates may be used to reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **802**.

[0160] The drain structure of HEMT **802** includes drain electrical contact **850** that extends through dielectric layer **860** and barrier layer **820** and contacts the drain region of channel layer **810**. In

addition to drain electrical contact **850**, the drain structure of HEMT **802** also includes a p-type drain contact structure that includes drain semiconductor layer **852** (e.g., a p-doped GaN layer) over the upper surface of barrier layer **820**. Drain semiconductor layer **852** may be separate from drain electrical contact **850**, or may contact drain electrical contact **850**. Drain semiconductor layer **852** may be fabricated by, for example, epitaxially growing a p-doped semiconductor layer over barrier layer **820** and selectively etching the p-doped semiconductor layer using an etch mask that defines the shape and size of drain semiconductor layer **852**. A drain metal contact is deposited on drain semiconductor layer **852**. The drain metal contact may be electrically connected to drain electrical contact **850** and/or a drain drive circuit.

[0161] The drain metal contact may be electrically connected to drain electrical contact **850**. The drain metal contact includes first metal portion **854**, second metal portion **856**, and third metal portion **858**. Second metal portion **856** is electrically connected to drain electrical contact **850** and may, in some examples, have the same material as drain electrical contact **850** and contact drain electrical contact **850**. First metal portion **854** is on a first semiconductor region of drain semiconductor layer **852**. Second metal portion **856** is on a second semiconductor region of drain semiconductor layer **852**. Third metal portion **858** is on a third semiconductor region of drain semiconductor layer **852**. First metal portion **854** and the first semiconductor region of drain semiconductor layer **852** may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the p-type drain contact structure, and may be used to inject holes to the barrier layer, the channel layer, and the buffer layer to neutralize the trapped electrons at the off-state and during the switching, thereby reducing the dynamic on-state resistance and current collapse. Second metal portion **856** and the second semiconductor region of drain semiconductor layer **852** may form a second junction that has a lower barrier height, such as an ohmic junction. Third metal portion **858** and the third semiconductor region of drain semiconductor layer **852** may form a third junction that has a barrier height between the first junction and the second junction, such as a low-barrier Schottky junction. The second junction and the third junction can provide a low-resistance connection to the p-GaN layer of the p-type drain contact structure to avoid the voltage dividing caused by the first junction, thereby increasing the voltage level at the p-GaN layer and thus the hole injection efficiency of the p-type drain contact structure. First metal portion **854**, second metal portion **856**, and third metal portion **858** may include different metal or metal alloy materials that have different work functions to achieve desired barrier heights, and may include materials described above with respect to FIG. 5A.

[0162] FIGS. 6A-8B illustrate examples of a hybrid drain contact structure that includes at least a high barrier junction and a lower barrier junction, and optionally one or more high or low barrier junctions. The arrangements of the junctions shown in FIGS. 6A-8B are for illustration purposes only. In other examples, the two or more metal-to-semiconductor junctions having different energy barrier heights may be arranged differently. In addition, the hybrid gate contact structures in enhancement mode HEMTs may also include two or more metal-to-semiconductor junctions having different energy barrier heights, where the hybrid gate contact structure and the hybrid drain contact structure may have the same or different numbers of metal-to-semiconductor junctions, the same or different metal-to-semiconductor junctions, and the same or different arrangements of the two or more metal-to-semiconductor junctions.

[0163] FIGS. 9A-9C illustrate an example of a drain region **900** of an HEMT including a hybrid drain contact structure according to certain examples. FIG. 9A is a top view of the hybrid drain contact structure at drain region **900**, FIG. 9B is a cross-sectional view of drain region **900** along a line AA', and FIG. 9C is a cross-sectional view of drain region **900** along a line BB'. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **500**, **600**, or **602**. In some examples, the hybrid drain contact structure shown in FIGS. 9A-9C may be a common hybrid drain contact structure shared by two half-pitches of an HEMT

device.

[0164] As shown in FIGS. 9A and 9B, the hybrid drain contact structure may include a plurality of drain contact structures **932** formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **910** (e.g., including GaN) and a barrier layer **920** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. Each of the plurality of drain contact structures **932** includes a semiconductor block (e.g., a p-GaN block) of a semiconductor layer **930** over barrier layer **920**, and a first metal portion **942** and a second metal portion **944** on semiconductor layer **930**. As described above, semiconductor layer **930** may deplete the 2DEG channel underneath. In the illustrated example, each semiconductor block has a hexagonal shape and tapers in a direction towards the gate structure of the HEMT. In other examples, each semiconductor block can have any other suitable regular, irregular, or arbitrary shapes, and can taper linearly or nonlinearly. As described above, the semiconductor blocks with the tapered shape may facilitate the movement of carriers in regions of the 2DEG channel between drain contact structures **932** and reduce the channel width reduction, thereby reducing the on-state resistance of the HEMT. First metal portion **942** and second metal portion **944** may include metal or metal alloy materials described above. First metal portion **942** and a first semiconductor portion of semiconductor layer **930** form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **944** and a second semiconductor portion of semiconductor layer **930** form a second junction (e.g., an ohmic junction) having a second energy barrier height lower than the first energy barrier height. In some examples, each drain contact structure **932** may also include a drain electrical contact **940** that contacts channel layer **910** to form an ohmic contact with channel layer **910**. In some examples, drain electrical contacts **940** may not be part of drain contact structures **932** and may be between drain contact structures **932**. In some examples, drain electrical contacts **940** and second metal portions **944** of the plurality of drain contact structures **932** may include the same metal material and may be deposited in a same deposition process. As shown in FIGS. 9A and 9C, second metal portions **944** of the plurality of drain contact structures **932** and drain electrical contacts **940** may be contiguous (adjacent and touching) and may be parts of a continuous metal contact structure.

[0165] FIGS. 9D-9E illustrate an example of a drain region **950** of an HEMT including a hybrid drain contact structure according to certain examples. FIG. 9D is a top view of the hybrid drain contact structure at drain region **950**, and FIG. 9E is a cross-sectional view of drain region **950** along a line CC'. A cross-sectional view of drain region **950** along a line DD' may be similar to the cross-sectional view shown in FIG. 9C. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **800** or **802**. In some examples, the hybrid drain contact structure shown in FIGS. 9D-9E may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0166] As shown in FIGS. 9D and 9E, the hybrid drain contact structure may include a plurality of drain contact structures **955** formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **960** (e.g., including GaN) and a barrier layer **970** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. Each of the plurality of drain contact structures **955** includes a semiconductor block (e.g., a p-GaN block) of a semiconductor layer **980** over barrier layer **970**, and a first metal portion **992**, a second metal portion **994**, and a third metal portion **996** on the semiconductor block. As described above, semiconductor layer **980** may deplete the 2DEG channel underneath. In the illustrated example, each semiconductor block has a hexagonal shape and tapers in a direction towards the gate structure of the HEMT. The semiconductor blocks with the tapered shape may facilitate the movement of carriers in regions of the 2DEG channel between drain contact structures **955** and reduce the channel width reduction, thereby reducing the on-state resistance of the HEMT. First metal portion **992**, second metal portion **994**, and third metal portion **996** may include metal or metal alloy materials described above. First metal portion **992** and a first semiconductor portion of

the semiconductor block form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **994** and a second semiconductor portion of the semiconductor block form a second junction (e.g., an ohmic junction) having a second energy barrier height lower than the first energy barrier height. Third metal portion **996** and a third semiconductor portion of the semiconductor block form a third junction (e.g., a low-barrier Schottky junction) having a third energy barrier height between the first energy barrier height and the second energy barrier height. In some examples, each drain contact structure **955** may also include a drain electrical contact **990** that contacts channel layer **960** to form an ohmic contact with channel layer **960**. In some examples, drain electrical contacts **990** may not be part of drain contact structures **955** and may be between drain contact structures **955**. In some examples, drain electrical contacts **990** and second metal portions **994** of the plurality of drain contact structures **955** may include the same metal material and may be deposited in a same deposition process. As shown in FIG. **9D** (and FIG. **9C**), second metal portions **994** of the plurality of drain contact structures **955** and drain electrical contacts **990** may be contiguous (adjacent and touching) and may be parts of a continuous metal contact structure.

[0167] FIGS. **10A-10C** illustrate an example of a drain region **1000** of an HEMT including a hybrid drain contact structure according to certain examples. FIG. **10A** is a top view of the hybrid drain contact structure at drain region **1000**, FIG. **10B** is a cross-sectional view of drain region **1000** of the HEMT along a line EE', and FIG. **10C** is a cross-sectional view of drain region **1000** of the HEMT along a line FF'. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **700** or **702**. In some examples, the hybrid drain contact structure shown in FIGS. **10A-10C** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0168] As shown in FIGS. **10A** and **10B**, the hybrid drain contact structure may include a plurality of drain contact structures **1032** formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1010** (e.g., including GaN) and a barrier layer **1020** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. Each of the plurality of drain contact structures **1032** includes a semiconductor block (e.g., a p-GaN block) of a semiconductor layer **1030** over barrier layer **1020**, and a first metal portion **1042** and a second metal portion **1044** on semiconductor layer **1030**. As described above, semiconductor layer **1030** may deplete the 2DEG channel underneath. In the illustrated example, each semiconductor block has a hexagonal shape and tapers in a direction towards the gate structure of the HEMT. As described above, the semiconductor blocks with the tapered shape may facilitate the movement of carriers in regions of the 2DEG channel between drain contact structures **1032**, thereby reducing the channel width reduction caused by the semiconductor blocks and reducing the on-state resistance of the HEMT. First metal portion **1042** and second metal portion **1044** may include metal or metal alloy materials described above. First metal portion **1042** and a first semiconductor portion of each semiconductor block form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **1044** and a second semiconductor portion of the semiconductor block form a second junction (e.g., a low-barrier Schottky junction) having a second energy barrier height lower than the first energy barrier height. In some examples, each drain contact structure **1032** may also include a drain electrical contact **1040** that contacts channel layer **1010** to form an ohmic contact with channel layer **1010**. In some examples, drain electrical contacts **1040** may not be part of drain contact structures **1032** and may be between drain contact structures **1032**. Drain electrical contacts **1040** and second metal portions **1044** of the plurality of drain contact structures **1032** may be contiguous (adjacent and touching) and may be parts of a continuous metal contact structure.

[0169] FIGS. **10D-10E** illustrate an example of a drain region **1050** of an HEMT including a hybrid drain contact structure according to certain examples. FIG. **10D** is a top view of the hybrid drain contact structure at drain region **1050**, and FIG. **10E** is a cross-sectional view of drain region **1050**

along a line GG'. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **700** or **702**. In some examples, the hybrid drain contact structure shown in FIGS. **10D-10E** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0170] As shown in FIGS. **10D** and **10E**, the hybrid drain contact structure may include a plurality of drain contact structures **1055** formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1060** (e.g., including GaN) and a barrier layer **1070** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. Each of the plurality of drain contact structures **1055** includes a semiconductor block (e.g., a p-GaN block) of a semiconductor layer **1080** over barrier layer **1070**, and a first metal portion **1092** and a second metal portion **1094** on the semiconductor block. As described above, semiconductor layer **1080** may deplete the 2DEG channel underneath. In the illustrated example, each semiconductor block has a hexagonal shape and tapers in a direction towards the gate structure of the HEMT. The semiconductor blocks with the tapered shape may facilitate the movement of carriers in regions of the 2DEG channel between drain contact structures **1055**, thereby reducing the channel width reduction caused by the semiconductor blocks and reducing the on-state resistance of the HEMT. First metal portion **1092** and second metal portion **1094** may include metal or metal alloy materials described above. First metal portion **1092** and a first semiconductor portion of each semiconductor block form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **1094** and a second semiconductor portion of the semiconductor block form a second junction (e.g., an LBS junction) having a second energy barrier height lower than the first energy barrier height.

[0171] The hybrid drain contact structure also include a drain electrical contact **1090** that contacts channel layer **1060** to form an ohmic contact with channel layer **1060**. In some examples, drain electrical contact **1090** may be a continuous metal contact structure, and may be formed by, for example, selectively etching semiconductor layer **1080** and barrier layer **1070** and depositing the metal material of drain electrical contact **1090** in the etched regions.

[0172] In some of the examples described above, the hybrid drain contact structure may include a plurality of drain contact structures formed using semiconductor blocks that are separate from each other. The layout of the hybrid drain contact structure may be complex, and the individual semiconductor blocks may need to be fabricated by epitaxial growth and etching using masks that may need to have a complex layout. In addition, in order to reduce the channel width reduction caused by the depletion of the 2DEG by the semiconductor blocks, each semiconductor block may need to be small, and the total area of the semiconductor blocks may need to be small. As result, the total area of the hybrid drain contact structure may be small, and thus the hole injection capability of the hybrid drain contact structure may not be as high as desired.

[0173] In some examples, in order to simplify the layout and increase the area of the hybrid drain contact structure, the hybrid drain contact structure of an HEMT may include a continuous drain electrical contact, and a p-type drain contact structure that includes a continuous p-type semiconductor region, a continuous first metal portion, and a second metal portion on the p-type semiconductor region. The drain electrical contact may contact the channel layer to form an ohmic contact with the channel layer. In some examples, the second metal portion and the drain electrical contact may be contiguous or may be parts of a same metal layer. In examples where the HEMT is an enhancement mode HEMT, the barrier layer under the hybrid drain contact structure may have a higher thickness than the barrier layer under the gate structure of the enhancement mode HEMT, and/or the p-GaN layer in the gate structure may have a higher thickness than the p-GaN layer in the hybrid drain contact structure, such that the 2DEG under the gate structure may be depleted, but the 2DEG under the hybrid drain contact structure may not be depleted and thus the on-state resistance can be low.

[0174] The different thicknesses of the barrier layer may be achieved by, for example, uniformly

growing a first sublayer of the barrier layer followed by selective area growth of a second sublayer of the barrier layer at the drain region. In another example, a barrier layer with a uniform thickness may be grown first and then selectively etched at the gate region to reduce the thickness of the barrier layer at the gate region. After the etching, a p-GaN layer may be grown on the partially etched barrier layer. The p-GaN layer may have a flat top surface such that the p-GaN layer at the gate region may be thicker (due to the lower thickness of the underlying barrier layer) than the p-GaN layer at the drain region. As such, at the gate region, the barrier layer may be thinner and the p-GaN layer may be thicker, such that the 2DEG at the gate region may be depleted, whereas the barrier layer may be thicker and the p-GaN layer may be thinner at the drain region, such that the 2DEG at the drain region may not be depleted. In some examples, additionally or alternatively, the sublayers of the barrier layer may have different compositions (e.g., different percentages of aluminum) in order to deplete the 2DEG at the gate region but not deplete the 2DEG at the drain region.

[0175] In some examples, the barrier layer may have a uniform thickness, but the p-GaN layer may have a higher thickness at the gate region than at the drain region. The different thicknesses of the p-GaN layer may be achieved by, for example, uniformly growing a first sublayer of the p-GaN layer followed by selective area growth of a second sublayer of the p-GaN layer at the gate region. In another example, a p-GaN layer with a uniform thickness may be grown first and then selectively etched at the drain region to reduce the thickness of the p-GaN layer at the drain region.

[0176] FIGS. **11A-11H** illustrate examples of hybrid drain contact structures of HEMTs including a continuous semiconductor layer (e.g., a p-GaN layer) according to certain examples. FIG. **11A** is a top view of an example of a hybrid drain contact structure at a drain region **1100** of an HEMT, and FIG. **11B** is a cross-sectional view of an example of drain region **1100** of the HEMT along a line II'. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **500**, **600**, or **602**. In some examples, the hybrid drain contact structure shown in FIGS. **11A-11B** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0177] As shown in FIGS. **11A** and **11B**, the hybrid drain contact structure may be formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1102** (e.g., including GaN) and a barrier layer **1104** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. The hybrid drain contact structure may include a semiconductor layer **1106** (e.g., a p-GaN layer) over barrier layer **1104**, and a first metal portion **1112** and a second metal portion **1114** on semiconductor layer **1106**. Semiconductor layer **1106** may extend along the channel width direction (e.g., the y direction). As described above, semiconductor layer **1106** may deplete the 2DEG channel underneath. To prevent semiconductor layer **1106** from completely depleting the 2DEG channel underneath, semiconductor layer **1106** may need to be thin, and/or barrier layer **1104** may need to be thick. In this way, the 2DEG channel under semiconductor layer may be conductive when the HEMT is in the Off state. On the other hand, at the gate region, semiconductor layer **1106** may need to be thick and/or barrier layer **1104** may need to be thin, such that the 2DEG channel under semiconductor layer **1106** may be depleted to form an enhancement mode HEMT. The different thicknesses of barrier layer **1104** and/or the semiconductor layer **1106** may be achieved by multiple epitaxial growth and/or selective etching processes described above.

[0178] First metal portion **1112** and second metal portion **1114** may include metal or metal alloy materials described above. First metal portion **1112** and a first semiconductor portion of semiconductor layer **1106** form a first junction (e.g., a Schottky junction) having a first energy barrier height. First metal portion **1112** may extend along the channel width direction (e.g., y direction). Second metal portion **1114** and a second semiconductor portion of semiconductor layer **1106** form a second junction (e.g., an ohmic junction) having a second energy barrier height lower than the first energy barrier height. Second metal portion **1114** may extend along the channel width

direction (e.g., the y direction). In some examples, semiconductor layer **1106** and barrier layer **1104** in some regions may be etched, and a drain electrical contact **1110** may be deposited to contact channel layer **1102** and form an ohmic contact with channel layer **1102**. Drain electrical contact **1110** may extend in the channel width direction (e.g., y direction). In some examples, second metal portion **1114** is a part of drain electrical contact **1110** deposited on semiconductor layer **1106** and form the second junction with semiconductor layer **1106**.

[0179] FIG. **11C** is a top view of an example of a hybrid drain contact structure at a drain region **1120** of an HEMT, and FIG. **11D** is a cross-sectional view of an example of drain region **1120** of the HEMT along line II'. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT, and may be an example of HEMT **700** or **702**. In some examples, the hybrid drain contact structure shown in FIGS. **11C-11D** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0180] As shown in FIGS. **11C** and **11D**, the hybrid drain contact structure may be formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1122** (e.g., including GaN) and a barrier layer **1124** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. The hybrid drain contact structure may include a semiconductor layer **1126** (e.g., a p-GaN layer) over barrier layer **1124**, and a first metal portion **1132** and a second metal portion **1134** on semiconductor layer **1126**. Semiconductor layer **1126** may extend along the channel width direction (e.g., the y direction). As described above, semiconductor layer **1126** may otherwise deplete the 2DEG channel underneath. To prevent semiconductor layer **1126** from completely depleting the 2DEG channel underneath, semiconductor layer **1126** may need to be thin, and/or barrier layer **1124** may need to be thick. In this way, the 2DEG channel under semiconductor layer **1126** may be conductive (on) when the HEMT is in the off state. On the other hand, at the gate region, semiconductor layer **1126** may need to be thick and/or barrier layer **1124** may need to be thin, such that the 2DEG channel under semiconductor layer **1126** may be depleted to form an enhancement mode HEMT. The different thicknesses of barrier layer **1124** and/or semiconductor layer **1126** at different regions may be achieved by multiple epitaxial growth and/or selective etching processes described above.

[0181] First metal portion **1132** and second metal portion **1134** may include metal or metal alloy materials described above, and may extend along the channel width direction (e.g., the y direction). First metal portion **1132** and a first semiconductor portion of semiconductor layer **1126** form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **1134** and a second semiconductor portion of semiconductor layer **1126** form a second junction (e.g., a low-barrier Schottky junction) having a second energy barrier height lower than the first energy barrier height. In the illustrated example, semiconductor layer **1126** and barrier layer **1124** in the middle region of the hybrid drain contact structure are etched, and a drain electrical contact **1130** is deposited to contact channel layer **1122** and form an ohmic contact with channel layer **1122**. Drain electrical contact **1130** may extend in the channel width direction (e.g., y direction). In the illustrated example, second metal portion **1134** is between first metal portion **1132** and drain electrical contact **1130**. In some examples, second metal portion **1134** may contact drain electrical contact **1130**.

[0182] FIG. **11E** is a top view of an example of a hybrid drain contact structure at a drain region **1140** of an HEMT, and FIG. **11F** is a cross-sectional view of an example of drain region **1140** of the HEMT along line II'. A cross-sectional view of drain region **1140** of the HEMT along a line JJ' may be similar to the cross-sectional view shown in FIG. **11D**. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT. In some examples, the hybrid drain contact structure shown in FIGS. **11E-11F** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0183] As shown in FIGS. **11E** and **11F**, the hybrid drain contact structure may be formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1142** (e.g.,

including GaN) and a barrier layer **1144** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. The hybrid drain contact structure may include a semiconductor layer **1146** (e.g., a p-GaN layer) over barrier layer **1144**, and a first metal portion **1152** and a second metal portion **1154** on semiconductor layer **1146**. Semiconductor layer **1146** may extend along the channel width direction (e.g., the y direction). To prevent semiconductor layer **1146** from completely depleting the 2DEG channel underneath, semiconductor layer **1146** may need to be thin, and/or barrier layer **1144** may need to be thick. In this way, the 2DEG channel under semiconductor layer **1126** may be conductive when the HEMT is in the off state. On the other hand, at the gate region, semiconductor layer **1146** may need to be thick and/or barrier layer **1144** may need to be thin, such that the 2DEG channel under semiconductor layer **1146** may be depleted to form an enhancement mode HEMT. The different thicknesses of barrier layer **1144** and/or the semiconductor layer **1146** at different regions may be achieved by multiple epitaxial growth and/or selective etching processes described above.

[0184] First metal portion **1152** and second metal portion **1154** may include metal or metal alloy materials described above and may also extend along the channel width direction (e.g., y direction). In the illustrated example, one or more regions **1156** of second metal portion **1154** are separated by first metal portion **1152**. First metal portion **1152** and a first semiconductor portion of semiconductor layer **1146** form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **1154** and a second semiconductor portion of semiconductor layer **1146** form a second junction (e.g., a low-barrier Schottky junction) having a second energy barrier height lower than the first energy barrier height. In the illustrated example, semiconductor layer **1146** and barrier layer **1144** in the middle region of the hybrid drain contact structure are etched, and a drain electrical contact **1150** is deposited in the etch regions to contact channel layer **1142** and form an ohmic contact with channel layer **1142**. Drain electrical contact **1150** may extend in the channel width direction (e.g., y direction). In some examples, second metal portion **1134** may contact drain electrical contact **1130**.

[0185] FIG. **11G** is a top view of an example of a hybrid drain contact structure at a drain region **1160** of an HEMT, and FIG. **11H** is a cross-sectional view of an example of drain region **1160** of the HEMT along line II'. A cross-sectional view of drain region **1160** of the HEMT along a line KK' may be similar to the cross-sectional view shown in FIG. **11D**. The HEMT may be an enhancement mode HEMT or a depletion mode HEMT. In some examples, the hybrid drain contact structure shown in FIGS. **11G-11H** may be a common hybrid drain contact structure shared by two half-pitches of an HEMT device.

[0186] As shown in FIGS. **11G** and **11H**, the hybrid drain contact structure may be formed on a semiconductor substrate. The semiconductor substrate may include a channel layer **1162** (e.g., including GaN) and a barrier layer **1164** (e.g., including AlGaN) grown on a growth substrate that may include one or more buffer layers. The hybrid drain contact structure may include a semiconductor layer **1166** (e.g., a p-GaN layer) over barrier layer **1164**, and a first metal portion **1172** and a second metal portion **1174** on semiconductor layer **1166**. Semiconductor layer **1166** may extend along the channel width direction (e.g., the y direction). To prevent semiconductor layer **1166** from completely depleting the 2DEG channel underneath, semiconductor layer **1166** may need to be thin, and/or barrier layer **1164** may need to be thick. In this way, the 2DEG channel under semiconductor layer **1166** may be conductive when the HEMT is in the off state. On the other hand, at the gate region, semiconductor layer **1166** may need to be thick and/or barrier layer **1164** may need to be thin, such that the 2DEG channel under semiconductor layer **1166** may be depleted to form an enhancement mode HEMT. The different thicknesses of barrier layer **1164** and/or the semiconductor layer **1166** at different regions may be achieved by multiple epitaxial growth and/or selective etching processes described above.

[0187] First metal portion **1172** and second metal portion **1174** may include metal or metal alloy materials described above, and may extend along the channel width direction (e.g., the y direction).

First metal portion **1172** and a first semiconductor portion of semiconductor layer **1166** form a first junction (e.g., a Schottky junction) having a first energy barrier height. Second metal portion **1174** and a second semiconductor portion of semiconductor layer **1166** form a second junction (e.g., a low-barrier Schottky junction) having a second energy barrier height lower than the first energy barrier height. In the illustrated example, semiconductor layer **1166** and barrier layer **1164** in some regions of the hybrid drain contact structure are etched, and drain electrical contacts **1170** are deposited to contact channel layer **1162** and form ohmic contacts with channel layer **1162**. In the illustrated example, drain electrical contacts **1170** are separated by second metal portion **1174**. In some examples, second metal portion **1174** may contact drain electrical contacts **1170** and may form a continuous metal structure with drain electrical contacts **1170**.

[0188] In some examples of the hybrid drain contact structure, the two or more junctions having different energy barrier heights may be formed using a same metal material and a same base semiconductor material (e.g., p-GaN) with different additional constituents (e.g., Al or In), different concentrations of an additional constituent (e.g., Al or In), different doping densities, and/or different levels of dopant activation in two or more different regions. In one example, the p-dopants (e.g., Mg) in the drain semiconductor layer (e.g., a p-GaN layer) may have different activation levels in different regions, for example, via selective hydrogen-based deactivation, such that the different regions may have different effective doping densities. In another example, the drain semiconductor layer may have different compositions in different regions, such as different concentrations of Al and/or In in a GaN layer. The drain semiconductor layer with the different doping densities, dopant activation levels, or material compositions at different regions may have different energy band structures, and thus may form junctions having different barrier heights with a same metal material or different metal materials. In some examples, a same metal material may be used in a hybrid drain contact structure to form an ohmic contact with the channel layer, form an ohmic or LBS junction with a first semiconductor region of the drain semiconductor layer, and form a Schottky junction with a second semiconductor region of the drain semiconductor layer.

[0189] FIG. **12A** illustrates an example of an enhancement mode HEMT **1200** including a hybrid drain contact structure having a Schottky metal and a semiconductor layer with different energy band structures at different regions according to certain examples. In the illustrated example, HEMT **1200** includes a substrate (not shown), a channel layer **1210** (e.g., including a GaN layer), a barrier layer **1220** (e.g., including an AlGaN layer), and drain, source, and gate structures. The drain, source, and gate structures are isolated by the dielectric material of a dielectric layer **1260**. The gate structure is between the drain structure and the source structure, and may be closer to the source structure. In some examples, HEMT **1200** shown in FIG. **12A** may be a half pitch of an HEMT device that includes HEMT **1200** and a mirrored version of HEMT **1200** that shares the drain structure with HEMT **1200**.

[0190] The substrate of HEMT **1200** may be similar to substrate **505** described above, and thus is not described in detail with respect to FIG. **12A**. Channel layer **1210** and barrier layer **1220** may be epitaxially grown on or in the substrate to form a heterostructure that may induce a 2DEG **1212** near the interface between channel layer **1210** and barrier layer **1220** due to the different energy band structures of channel layer **1210** and barrier layer **1220**. Channel layer **1210**, barrier layer **1220**, and 2DEG **1212** may be similar to Channel layer **510**, barrier layer **520**, and 2DEG **512**, respectively. In the illustrated example, channel layer **1210** includes a gallium nitride (GaN) layer, and barrier layer **1220** includes an aluminum gallium nitride (AlGaN) layer. Other materials may also be used for channel layer **1210** and barrier layer **1220** as described above with respect to FIG. **5A**.

[0191] The gate structure of HEMT **1200** includes a gate semiconductor layer over an upper surface of barrier layer **1220**. The gate semiconductor layer may be similar to gate semiconductor layer **530** and includes a p-doped semiconductor layer, such as a gallium nitride (GaN) layer doped with magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof, as described

above with respect to FIG. 5A. The gate semiconductor layer may include a first region **1230** and a second region **1232** that have different energy band structures. As discussed above, first region **1230** and second region **1232** of the gate semiconductor layer may include the same base material, such as GaN, but may have different additional constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation, and thus may have different energy band structures. The p-doping density and the thickness of each region of the gate semiconductor layer and the thickness of barrier layer **1220** under the gate semiconductor layer may be selected such that the gate semiconductor layer may deplete 2DEG **1212** under the gate semiconductor layer, such that HEMT **1200** is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0192] A gate electrical contact **1234** is deposited on the gate semiconductor layer to function as a gate electrode for applying a gate voltage to the gate semiconductor layer. Gate electrical contact **1234** may include a Schottky metal. Gate electrical contact **1234** and first region **1230** of the gate semiconductor layer may form a Schottky contact, which may reduce gate leakage. Gate electrical contact **1234** and second region **1232** of the gate semiconductor layer may form an ohmic contact or a low-barrier Schottky contact, which may reduce the voltage drop between gate electrical contact **1234** and the gate semiconductor layer under DC conditions and in switching events, thereby reducing the effective gate overdrive voltage. In the illustrated example, the gate structure also includes a gate field plate **1236**. Gate field plate **1236** may be used to, for example, reduce current collapse and dynamic on-state resistance, and increase the breakdown voltage of HEMT **1200**.

[0193] At the source region of HEMT **1200**, a source electrical contact **1240** extends through barrier layer **1220** and contacts a source region of channel layer **1210**. Source electrical contact **1240** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **1210**. One or more source field plates **1242** and **1244** may be formed in dielectric layer **1260** and may be coupled to source electrical contact **1240**. As described above, the field plates may be used to reduce current collapse and dynamic on-state resistance and/or increase the breakdown voltage of HEMT **1200**.

[0194] At the drain region of HEMT **1200**, a drain electrical contact **1250** extends through dielectric layer **1260** and barrier layer **1220** and contacts a drain region of channel layer **1210**. Drain electrical contact **1250** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **1210**. In addition to drain electrical contact **1250**, the drain structure of HEMT **1200** also includes a p-type drain contact structure that includes a drain semiconductor layer (e.g., a p-doped GaN layer) over the upper surface of barrier layer **1220**. The drain semiconductor layer may include a first region **1252** and a second region **1254** that have different energy band structures. As discussed above, first region **1252** and second region **1254** of the drain semiconductor layer may include the same base material, such as GaN, but may have different additional constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation, and thus may have different energy band structures. In some examples, first region **1230** of the gate semiconductor layer and first region **1252** of the drain semiconductor layer include the same material and have the same energy band structure and the same thickness, and may be fabricated using the same processes. In some examples, second region **1232** of the gate semiconductor layer and second region **1254** of the drain semiconductor layer include the same material and have the same energy band structure and the same thickness, and may be fabricated using the same processes. The drain semiconductor layer may be separate from drain electrical contact **1250**, or may contact drain electrical contact **1250**.

[0195] In the illustrated example, a drain metal contact **1256** is deposited on the drain semiconductor layer. Drain metal contact **1256** may include the same metal material as gate electrical contact **1234**. Drain metal contact **1256** may be electrically connected to drain electrical contact **1250**, but may have a material having a metal work function different from the metal work

function of drain electrical contact **1250**. Drain metal contact **1256** and first region **1252** of the drain semiconductor layer may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the hybrid drain contact structure and may have some hole injection and current collapse mitigation capability at high drain voltages. Drain metal contact **1256** and second region **1254** of the drain semiconductor layer may form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction. The second junction with the lower barrier height can provide a Kelvin connection to the drain semiconductor layer of the hybrid drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height, thereby increasing the voltage level at the drain semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure. Drain metal contact **1256** may include, for example, titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), or an alloy thereof. In some examples, the alloy may include titanium tungsten aluminum (TiWAl) or titanium aluminum nitride (TiAlN).

[0196] FIG. **12B** illustrates an example of an enhancement mode HEMT **1202** including a hybrid drain contact structure having an ohmic metal and a semiconductor layer with different energy band structures at different regions according to certain examples. HEMT **1202** shown in FIG. **12B** may be similar to HEMT **1200** of FIG. **12A**, except the gate structure and the drain structure. In HEMT **1202**, the gate structure includes a gate semiconductor layer over an upper surface of barrier layer **1220**. The gate semiconductor layer may include a p-doped semiconductor layer, such as a gallium nitride (GaN) layer doped with magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof, as described above with respect to FIG. **5A**. The gate semiconductor layer may include a first region **1270** and a second region **1272** that have different energy band structures. As discussed above, first region **1270** and second region **1272** of the gate semiconductor layer may include the same base material, such as GaN, but may have different additional constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation, and thus may have different energy band structures. The p-doping density and the thickness of each region of the gate semiconductor layer and the thickness of barrier layer **1220** under the gate semiconductor layer may be selected such that the gate semiconductor layer may deplete a 2DEG **1214** under the gate semiconductor layer, such that HEMT **1202** is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure. In some examples, first region **1270** of the gate semiconductor layer of HEMT **1202** and first region **1230** of the gate semiconductor layer of HEMT **1200** may have different energy band structures, and second region **1272** of the gate semiconductor layer of HEMT **1202** and second region **1232** of the gate semiconductor layer of HEMT **1200** may have different energy band structures.

[0197] A gate electrical contact **1274** is deposited on the gate semiconductor layer to function as a gate electrode for applying a gate voltage to the gate semiconductor layer. Gate electrical contact **1274** may include an ohmic metal that may otherwise form an ohmic contact with a p-GaN material. Due to the different modifications to different regions of the gate semiconductor layer, gate electrical contact **1274** and first region **1270** of the gate semiconductor layer may form a Schottky contact, which may reduce gate leakage, whereas gate electrical contact **1274** and second region **1272** of the gate semiconductor layer may form an ohmic contact or a low-barrier Schottky contact, which may reduce the voltage drop between gate electrical contact **1274** and the gate semiconductor layer under DC conditions and in switching events, thereby reducing the effective gate overdrive voltage.

[0198] At the drain region of HEMT **1202**, drain electrical contact **1250** extends through dielectric layer **1260** and barrier layer **1220** and contacts a drain region of channel layer **1210**. Drain electrical contact **1250** may include a metal or metal alloy and may form a low-barrier metal-to-semiconductor contact (e.g., an ohmic contact) with channel layer **1210**. In addition to drain

electrical contact **1250**, the drain structure of HEMT **1200** also includes a p-type drain contact structure that includes a drain semiconductor layer (e.g., a p-doped GaN layer) over the upper surface of barrier layer **1220**. The drain semiconductor layer may include a first region **1280** and a second region **1282** that have different energy band structures as described above. In some examples, first region **1270** of the gate semiconductor layer and first region **1280** of the drain semiconductor layer include the same material and have the same energy band structure and the same thickness, and may be fabricated using the same processes. In some examples, second region **1232** of the gate semiconductor layer and second region **1282** of the drain semiconductor layer include the same material and have the same energy band structure and the same thickness, and may be fabricated using the same processes. The drain semiconductor layer may be separate from drain electrical contact **1250**, or may contact drain electrical contact **1250**.

[0199] In the illustrated example, a drain metal contact **1284** is deposited on the drain semiconductor layer. Drain metal contact **1284** may include the same metal material as gate electrical contact **1274**. Drain metal contact **1284** may be electrically connected to drain electrical contact **1250**, and may include the same material as drain electrical contact **1250**. Drain metal contact **1284** and first region **1280** of the drain semiconductor layer may form a first junction that has a higher barrier height, such as a Schottky junction. The first junction with the higher barrier height may reduce the leakage current of the hybrid drain contact structure and may have some hole injection and current collapse mitigation capability at high drain voltages. Drain metal contact **1284** and second region **1282** of the drain semiconductor layer may form a second junction that has a lower barrier height, such as an ohmic contact or a low-barrier Schottky junction. The second junction with the lower barrier height can provide a Kelvin connection to the drain semiconductor layer of the hybrid drain contact structure to avoid the voltage dividing caused by the first junction that has a higher barrier height as described above, thereby increasing the voltage level at the drain semiconductor layer and thus the hole injection efficiency of the p-type drain contact structure. Drain metal contact **1284** may include, for example, titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), or an alloy thereof. In some examples, the alloy may include titanium tungsten aluminum (TiWAl) or titanium aluminum nitride (TiAlN).

[0200] Even though FIGS. **12A** and **12B** describe examples of hybrid drain contact structures in enhancement mode HEMTs, the hybrid drain contact structures described with respect to FIGS. **12A** and **12B** can also be used in depletion mode HEMTs. In addition, the hybrid drain contact structures described with respect to FIGS. **12A** and **12B** can be used in HEMTs that have a plurality of p-GaN blocks (e.g., as described with respect to FIGS. **9A-10E**) or one continuous p-GaN block (e.g., as described with respect to FIGS. **11A-11H**) in the drain structure. Furthermore, the drain semiconductor layer can include two or more different regions having different energy band structures and thus can form two or more junctions having different barrier heights with a metal layer.

[0201] FIG. **12C** illustrates another example of an enhancement mode HEMT **1204** including a hybrid drain contact structure. HEMT **1204** may be similar to HEMT **1202**, but the gate semiconductor layer and the drain semiconductor layer in HEMT **1204** may each include three (or more) regions having different energy band structures. The three regions having different energy band structures may form three junctions having different barrier heights with a metal layer, such as a Schottky junction, a low-barrier Schottky junction, and an ohmic junction.

[0202] For example, the gate semiconductor layer in HEMT **1204** may include a first region **1262**, a second region **1264**, and a third region **1266** that have different energy band structures. As discussed above, in some examples, first region **1262**, second region **1264**, and third region **1266** of the gate semiconductor layer may include the same base material, such as GaN, but may have different additional constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation, and thus may have different energy band structures. First region **1262**, second

region **1264**, and third region **1266** may form three junctions having different barrier heights with a gate electrical contact **1268**, such as a Schottky junction, a low-barrier Schottky junction, and an ohmic junction.

[0203] Similarly, the drain semiconductor layer in HEMT **1204** may include a first region **1290**, a second region **1292**, and a third region **1294** that have different energy band structures. As discussed above, in some examples, first region **1290**, second region **1292**, and third region **1294** of the drain semiconductor layer may include the same base material, such as GaN, but may have different additional constituents (e.g., Al or In), different doping densities, and/or different levels of dopant activation, and thus may have different energy band structures. First region **1290**, second region **1292**, and third region **1294** may form three junctions having different barrier heights with a drain metal contact **1296**, such as a Schottky junction, a low-barrier Schottky junction, and an ohmic junction.

[0204] Even though FIGS. **12A-12C** show examples of HEMTs including the same number of regions having different energy band structures in the gate semiconductor layer and the drain semiconductor layer, the number of regions having different energy band structures in the gate semiconductor layer may be different from the number of regions having different energy band structures in the drain semiconductor layer in other examples.

[0205] FIGS. **13A-13D** illustrate an example of a process of fabricating an HEMT including a hybrid drain contact structure according to certain examples. FIG. **13A** shows a semiconductor substrate that includes a channel layer **1310** (e.g., including a GaN layer) grown on a growth substrate, and a barrier layer **1320** (e.g., including an AlGa_N) layer on channel layer **1310**. The semiconductor substrate also include a semiconductor layer grown on barrier layer **1320** and patterned to form a gate semiconductor layer **1330** and a drain semiconductor layer **1340**. A passivation layer **1350** (e.g., including an oxide or nitride based dielectric material) is deposited over barrier layer **1320**, gate semiconductor layer **1330**, and drain semiconductor layer **1340**, to encapsulate these semiconductor layers. As described above with respect to, for example, FIG. **5A**, the growth substrate may be a bulk semiconductor substrate, a semiconductor-on-insulator (SOI) substrate, or any other appropriate substrate. In one example, the growth substrate may include a bulk silicon wafer, and may include one or more transition layers or buffer layers of suitable materials for accommodating the lattice mismatch between the growth substrate and channel layer **1310** (e.g., to reduce or minimize lattice defect generation and/or propagation in channel layer **1310**). For example, the transition layers or buffer layers may have a gradient concentration of one or more elements in a surface normal direction (e.g., z direction) of the semiconductor substrate.

[0206] Channel layer **1310** and barrier layer **1320** may be epitaxially grown on the growth substrate to form a heterostructure that may induce a 2DEG **1312** near the interface between channel layer **1310** and barrier layer **1320** due to the different energy band structures of channel layer **1310** and barrier layer **1320**. 2DEG **1312** may conduct current in a two-dimensional plane (e.g., an x-y plane). In the illustrated example, channel layer **1310** includes a GaN layer. In some examples, the material of channel layer **1310** includes an unintentionally doped material, such as a material doped by diffusion of dopants from another layer, or includes an intrinsic material. In the illustrated example, barrier layer **1320** includes an aluminum gallium nitride (AlGa_N) layer. Other materials may be used for channel layer **1310** and barrier layer **1320**. For example, channel layer **1310** may include indium aluminum gallium nitride (In.sub.iAl.sub.jGa.sub.1-i-jN) (where 0≤i≤1, 0≤j≤1, and 0≤i+j≤1), and barrier layer **1320** may include indium aluminum gallium nitride (In.sub.kAl.sub.lGa.sub.1-k-lN) (where 0≤k≤1, 0≤l≤1, and 0≤k+l≤1).

[0207] Gate semiconductor layer **1330** and drain semiconductor layer **1340** are fabricated in a p-doped semiconductor layer. The p-doped semiconductor layer may have the same base material as channel layer **1310**. For example, the p-doped semiconductor layer may include a gallium nitride (Ga_N) layer, or more generally, indium aluminum gallium nitride (In.sub.mAl.sub.nGa.sub.1-m-nN) (where 0≤m<1, 0≤n<1, and 0≤m+n≤1). The p-type dopant of the p-doped semiconductor layer

may include magnesium (Mg), carbon (C), zinc (Zn), and the like, or a combination thereof. In some examples, a concentration of the p-type dopant that is electrically activated in the p-doped semiconductor layer may be equal to or greater than $1 \times 10^{17} \text{ cm}^{-3}$, or equal to or greater than $1 \times 10^{18} \text{ cm}^{-3}$. Other materials, dopants, and/or concentrations may be used in other examples. Gate semiconductor layer **1330** and drain semiconductor layer **1340** may be formed by epitaxial growth and selective etching using an etch mask, or may be formed by selective area growth using a growth mask. The etch mask or growth mask may define the shape and size of gate semiconductor layer **1330** and drain semiconductor layer **1340**. For example, as described above, drain semiconductor layer **1340** may have a hexagonal shape and may taper towards the gate structure. The doping density and thickness of p-doped gate semiconductor layer **1330** and the thickness of barrier layer **1320** under gate semiconductor layer **1330** may be selected such that the p-doped gate semiconductor layer **1330** may deplete 2DEG **1312** under gate semiconductor layer **1330**, and thus the HEMT is off without a positive gate voltage and may be turned on by applying a positive voltage to the gate structure.

[0208] As described above with respect to, for example, FIGS. **12A** and **12B**, in some examples, each of gate semiconductor layer **1330** and drain semiconductor layer **1340** may include a same base semiconductor material (e.g., p-GaN) with different additional constituents (e.g., Al or In), different concentrations of an additional constituent (e.g., Al or In), different doping densities, and/or different levels of dopant activation in two or more different regions. In one example, the p-dopants (e.g., Mg) in gate semiconductor layer **1330** and drain semiconductor layer **1340** may have different activation levels in different regions, for example, via selective hydrogen-based deactivation, such that the different regions may have different effective doping densities and different energy band structures. In another example, each of gate semiconductor layer **1330** and drain semiconductor layer **1340** may have different compositions in different regions, such as different concentrations of Al and/or In in a GaN layer. The drain semiconductor layer with the different doping densities, dopant activation levels, or material compositions at different regions may have different energy band structures, and thus may form two or more junctions having different barrier heights with a same metal material or different metal materials.

[0209] Passivation layer **1350** may include one or more dielectric layers of a same dielectric material or different dielectric materials deposited in one or more deposition processes. For example, passivation layer **1350** may include an oxide-based dielectric material or a nitride-based dielectric material, such as silicon oxide (e.g., a phosphosilicate glass (PSG)), aluminum oxide, silicon nitride, and the like.

[0210] FIG. **13B** shows that passivation layer **1350** and barrier layer **1320** may be selectively etched to expose channel layer **1310** at the source and drain regions and a first region of drain semiconductor layer **1340**, and then a first metal material may be deposited on the exposed channel layer **1310** and drain semiconductor layer **1340**. The first metal material deposited on the exposed channel layer at the source region is a source electrical contact **1360**, which may form an ohmic contact with channel layer **1310**. The first metal material deposited on the exposed channel layer at the drain region is a drain electrical contact **1362**, which may form an ohmic contact with channel layer **1310**. The first metal material deposited on the first region of drain semiconductor layer **1340** is a first drain metal contact **1364**, which may form an ohmic contact with the first region of drain semiconductor layer **1340**. In some examples, the first metal material includes titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), or an alloy thereof.

[0211] FIG. **13C** shows that passivation layer **1350** at the drain region and the gate region may be selectively etched to expose gate semiconductor layer **1330** and a second region of drain semiconductor layer **1340**, and then a second metal material is deposited on the exposed gate semiconductor layer **1330** and the second region of drain semiconductor layer **1340**. The second metal material deposited on the second region of drain semiconductor layer **1340** is a second drain

metal contact **1372**, which may form a Schottky contact with the second region of drain semiconductor layer **1340**. Therefore, there may be three types of metal-to-semiconductor contact at the drain region, including an ohmic contact between drain electrical contact **1362** and channel layer **1310**, an ohmic contact between first drain metal contact **1364** and the first region of drain semiconductor layer **1340**, and a Schottky contact between second drain metal contact **1372** and the second region of drain semiconductor layer **1340**. The second metal material deposited on the exposed gate semiconductor layer **1330** is a gate metal contact **1370**, which may form a Schottky contact with gate semiconductor layer **1330**. In some examples, the second metal material includes titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), or an alloy thereof.

[0212] FIG. **13D** shows additional structures fabricated using backend-of-line (BEOL) processes. For example, the deposited second metal material may be planarized, where the remaining second metal material on passivation layer **1350** and adjacent to gate metal contact **1370** may be a gate field plate **1380**. Additional dielectric material layers may be deposited, and metal layers may be deposited on the dielectric material layers and patterned to form, for example, a first source field plate **1382** and a second source field plate **1384**. As described above, the field plates may be used to, for example, mitigate the current collapse and the increase breakdown voltage of the HEMT. Even though not shown in FIG. **13D**, electrical interconnects, such as metal traces and vias, may be formed to connect the drain structure, gate structure, and source structure to drive circuits or load circuits.

[0213] FIGS. **13A-13D** show an example of processes for fabricating an example of an HEMT including an example of a hybrid drain contact structure disclosed herein according to certain examples. Other hybrid drain contact structures disclosed herein may be fabricated using similar processes, including various combinations of epitaxial growth, deposition, etching, planarization, and the like.

[0214] FIG. **14** includes a flowchart **1400** illustrating an example of a process of fabricating an HEMT including a hybrid drain contact structure according to certain examples. It is noted that the operations illustrated in FIG. **14** provide particular processes for fabricating examples of HEMTs disclosed herein according to certain examples. Other sequences of operations can also be performed to fabricate HEMTs according to alternative examples. For example, alternative examples may perform the operations in a different order. Moreover, the individual operations illustrated in FIG. **14** can include multiple sub-operations that can be performed in various sequences as appropriate for the individual operation. Furthermore, some operations can be added or removed depending on the particular example. In some examples, two or more operations may be performed in parallel. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

[0215] Operations at block **1410** of flowchart **1400** include forming a patterned semiconductor layer over a barrier layer on a semiconductor substrate. The barrier layer may be over a channel layer on the semiconductor substrate and may form a heterostructure with the channel layer such that 2DEG may be induced near the interface between the channel layer and the barrier layer. In one example, the channel layer includes a GaN layer, and the barrier layer includes an AlGaN layer. The patterned semiconductor layer may include a plurality of semiconductor regions. In some examples, each of the plurality of semiconductor regions includes at least a first semiconductor portion and a second semiconductor portion. The patterned semiconductor layer may be a p-doped semiconductor layer (e.g., a p-GaN layer) etched using an etch mask to selectively remove some regions of the p-doped semiconductor layer. The etch mask may define the shape and size of each of the plurality of semiconductor regions. Examples of the shape of each of the plurality of semiconductor regions are shown in, for example, FIGS. **5B, 9A, 9D, 10A, 10D, 11A, 11C, 11E,** and **11G**. For example, each of the plurality of semiconductor regions may have a hexagonal shape or another shape, and may taper in a direction from a drain region to a gate region. The patterned

semiconductor layer may include semiconductor regions at both the gate region and the drain region for fabricating an enhancement mode HEMT, or may only include semiconductor regions at the drain region for fabricating a depletion model HEMT.

[0216] In some examples, the semiconductor regions at the gate regions and the drain regions have different thicknesses, different dopants, different doping densities, different dopant activation levels, or different additional constituents. For example, the different thicknesses may be attained by a uniform epitaxial growth process and a selected area epitaxial growth process. The different activation levels in different regions can be attained by, for example, selective hydrogen-based deactivation. In some examples, there may be one semiconductor block at the drain region as described above with respect to, for example, FIGS. **11A-11H**. In some examples, the barrier layer may have different thicknesses at the gate region and the drain region, which may be achieved by, for example, a uniform epitaxial growth process and a selected area epitaxial growth process.

[0217] At block **1420**, the operations include depositing a passivation layer over the barrier layer and the patterned semiconductor layer. The passivation layer may include, for example, an oxide-based dielectric layer or a nitride-based dielectric layer. The passivation layer may be used to protect and isolate the source structure, gate structure, and drain structure. A semiconductor wafer formed after the operations of block **1420** is shown in FIG. **13A**.

[0218] Operations at block **1430** include selectively etching the passivation layer and the barrier layer to expose regions of the channel layer and first semiconductor portions of the plurality of semiconductor regions of the patterned semiconductor layer. For example, as described above with respect to FIG. **13B**, passivation layer **1350** and barrier layer **1320** may be selectively etched using one or more masks and one or more etching processes to expose channel layer **1310** at the source and drain regions and a first region of drain semiconductor layer **1340** at each p-type drain contact structure.

[0219] At block **1440**, first metal contacts are formed in the etched regions by depositing a first metal material on the exposed regions of the channel layer and the exposed first semiconductor portions of the plurality of semiconductor regions of the patterned semiconductor layer, as shown in, for example, FIG. **13B**. The first semiconductor portions and the first metal contacts on the first semiconductor portions form first junctions that may have low barrier heights, such as ohmic junctions or low-barrier Schottky junctions. The first metal contacts deposited on the exposed regions of the channel layer include one or more drain electrical contacts that contact the channel layer at the drain region and form low barrier junctions (e.g., ohmic junctions) with the channel layer. The first metal contacts deposited on the exposed regions of the channel layer also include one or more source electrical contacts that contact the channel layer at the source region and form low barrier junctions (e.g., ohmic junctions) with the channel layer.

[0220] Operations at block **1450** include removing the passivation layer on the second semiconductor portions of the plurality of semiconductor regions of the patterned semiconductor layer by etching to expose the second semiconductor portions of the plurality of semiconductor regions of the patterned semiconductor layer. In examples where enhancement mode HEMTs are fabricated, the passivation layer on the patterned semiconductor layer at the gate region may be etched to expose a semiconductor region of the patterned semiconductor layer at the gate region.

[0221] At block **1460**, second metal contacts may be formed by depositing a second metal material on the second semiconductor portions of the plurality of semiconductor regions of the patterned semiconductor layer as shown in, for example, FIG. **13C**. The second semiconductor portions and the second metal contacts on the second semiconductor portions form second junctions having energy barrier heights greater than the energy barrier heights of the first junctions. In examples where enhancement mode HEMTs are fabricated, the second metal contacts deposited on the patterned semiconductor layer at the gate region form third junctions with the patterned semiconductor layer at the gate region. The third junctions may have energy barrier heights greater than energy barrier heights of the first junctions, and may have energy barrier heights similar to the

energy barrier heights of the second junctions.

[0222] Even though not shown in flowchart **1400**, BEOL processes may be performed to fabricate additional structures, such as field plates, electrical interconnects, and the like. FIG. **14** shows an example of a process for fabricating an example of an HEMT including an example of a hybrid drain contact structure disclosed herein according to certain examples. Other hybrid drain contact structures disclosed herein may be fabricated using similar processes, including various combinations of epitaxial growth, deposition, etching, planarization, and the like. For example, even though a method of fabricating a hybrid drain contact structure with a plurality of p-type drain contact structures is described with respect to FIG. **14**, the method can be modified to include two or more epitaxial growth processes for growing the barrier layer or the p-type semiconductor layer, to fabricate a hybrid drain contact structure with one continuous p-type drain contact structure as described with respect to, for example, FIGS. **11A-11H**. In various examples, the HEMTs disclosed herein may be fabricated using, for example, a gate-first process flow, a gate-last process flow, or a hybrid process flow, and the gate and drain structures may be fabricated using a self-aligned process flow, a non-self-aligned process flow, or a hybrid alignment process flow.

[0223] As used herein, the term “contiguous” is used to describe adjacent structures that are connected without a break. As used herein, the term “semiconductor substrate” may refer to a semiconductor wafer without other layers or circuits formed therein, or a semiconductor wafer that includes various layers and circuits formed therein. As used herein, the term “layer” may refer to a continuous layer, or a region or portion of a layer. As used herein, terms “Schottky junction” and “Schottky contact” are used interchangeably. As used herein, terms “ohmic junction” and “ohmic contact” are used interchangeably. As used herein, terms “low-barrier Schottky junction” and “low-barrier Schottky contact” are used interchangeably.

[0224] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0225] Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0226] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0227] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0228] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for

example, by an end-user and/or a third-party.

[0229] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0230] References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

[0231] References herein to a FET being “on” or “enabled” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “off” or “disabled” means that the conduction channel is not present so drain current does not flow through the FET. An “off” FET, however, may have current flowing through the transistor's body-diode.

[0232] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0233] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other examples, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0234] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description.

[0235] In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0236] Terms “and” and “or,” as used herein, may include a variety of meanings that are also expected to depend at least in part upon the context in which such terms are used. Typically, “or” if used to associate a list, such as A, B, or C, is intended to mean A, B, and C, here used in the inclusive sense, as well as A, B, or C, here used in the exclusive sense. In addition, the term “one or more” as used herein may be used to describe any feature, structure, or characteristic in the singular or may be used to describe some combination of features, structures, or characteristics. However, it

should be noted that this is merely an illustrative example and claimed subject matter is not limited to this example. Furthermore, the term “at least one of” if used to associate a list, such as A, B, or C, can be interpreted to mean A, B, C, or a combination of A, B, and/or C, such as AB, AC, BC, AA, ABC, AAB, ACC, AABBBCCC, or the like.

[0237] Although various examples have been described in detail, it should be understood that various changes, substitutions, and alterations can be made therein without departing from the scope defined by the appended claims. The devices, structures, materials, and processes discussed above are examples. Various examples may omit, substitute, or add various procedures or components as appropriate. Also, features described with respect to certain examples may be combined in various other examples. Different aspects and elements of the examples may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples that do not limit the scope of the disclosure to those specific examples.

[0238] Specific details are given in the description on order to provide a thorough understanding of the examples. However, examples may be practiced without these specific details. For example, well-known circuits, processes, systems, structures, and techniques may have been shown without unnecessary detail in order to avoid obscuring the examples. This description provides examples only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the preceding description of the examples will provide those skilled in the art with an enabling description for implementing various examples. Various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the present disclosure. Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A semiconductor device comprising: a substrate including a channel layer and a barrier layer over the channel layer; a source contact electrically coupled to the channel layer; a gate structure over the barrier layer; and a drain contact structure including: a first drain contact electrically coupled to the channel layer; a semiconductor layer over the barrier layer and including a first semiconductor portion and a second semiconductor portion; and a second drain contact on the semiconductor layer and electrically coupled to the first drain contact, the second drain contact including a first metal portion and a second metal portion, the first metal portion and the first semiconductor portion forming a first junction having a first energy barrier height, and the second metal portion and the second semiconductor portion forming a second junction having a second energy barrier height lower than the first energy barrier height.
2. The semiconductor device of claim 1, wherein: the first junction includes a first Schottky junction; and the second junction includes an ohmic junction, a second Schottky junction having an energy barrier height lower than the first energy barrier height, or a combination thereof.
3. The semiconductor device of claim 1, wherein: the first metal portion includes a first metal material having a first metal work function; and the second metal portion includes a second metal material having a second metal work function different from the first metal work function.
4. The semiconductor device of claim 1, wherein: the first metal portion and the second metal portion include a same metal material; and the first semiconductor portion and the second semiconductor portion have different energy band structures.
5. The semiconductor device of claim 4, wherein the first semiconductor portion and the second semiconductor portion include a same base semiconductor material and have different levels of doping or dopant activation.
6. The semiconductor device of claim 1, wherein the first metal portion and the second metal portion include titanium (Ti), titanium tungsten (TiW), titanium nitride (TiN), nickel (Ni), platinum (Pt), tantalum nitride (TaN), copper (Cu), tungsten (W), gold (Au), aluminum (Al), titanium

tungsten aluminum (TiWAl), titanium aluminum nitride (TiAlN), or a combination thereof.

7. The semiconductor device of claim 1, wherein: the semiconductor layer further includes a third semiconductor portion; the second drain contact further includes a third metal portion; and the third semiconductor portion and the third metal portion form a third junction having a third energy barrier height different from the first energy barrier height and the second energy barrier height.
8. The semiconductor device of claim 7, wherein the third metal portion has a metal work function different from metal work functions of the first metal portion and the second metal portion.
9. The semiconductor device of claim 7, wherein the third semiconductor portion has an energy band structure different from energy band structures of the first semiconductor portion and the second semiconductor portion.
10. The semiconductor device of claim 1, wherein: the drain contact structure is a first drain contact structure; the semiconductor device includes a plurality of drain contact structures including the first drain contact structure, each of the plurality of drain contact structures having the same structure as the first drain contact structure; and the first drain contacts and the second metal portions of the second drain contacts of the plurality of drain contact structures are contiguous.
11. The semiconductor device of claim 10, wherein the semiconductor layer of the first drain contact structure is separate from the semiconductor layers of other drain contact structures of the plurality of drain contact structures.
12. The semiconductor device of claim 10, wherein the first semiconductor portion of the semiconductor layer and the first metal portion of the second drain contact of each drain contact structure of the plurality of drain contact structures are tapered in a direction towards the gate structure.
13. The semiconductor device of claim 10, wherein the first drain contacts of the plurality of drain contact structures are interleaved with the second metal portions of the second drain contacts and the semiconductor layers of the plurality of drain contact structures in a width direction of the semiconductor device.
14. The semiconductor device of claim 13, wherein the first drain contact and the second metal portion of the second drain contact of each of the plurality of drain contact structures have a same length in a length direction of the semiconductor device.
15. The semiconductor device of claim 13, wherein: the second drain contact of each of the plurality of drain contact structures further includes a third metal portion between the first metal portion and the second metal portion; the semiconductor layer of each of the plurality of drain contact structures further includes a third semiconductor portion; the third metal portion and the third semiconductor portion forms a third junction; and the third junction has a third energy barrier height different from the first energy barrier height and the second energy barrier height.
16. The semiconductor device of claim 13, wherein the first drain contact and the second metal portion of the second drain contact of each of the plurality of drain contact structures have different lengths in a length direction of the semiconductor device.
17. The semiconductor device of claim 10, wherein: the first drain contacts of the plurality of drain contact structures are contiguous; and the second metal portion of the second drain contact is between the first drain contact and the first metal portion of the second drain contact in each of the plurality of drain contact structures.
18. The semiconductor device of claim 1, wherein: the semiconductor layer includes a semiconductor region that extends continuously along a width direction of the semiconductor device; the first metal portion of the second drain contact extends along the width direction of the semiconductor device; and a first region of the barrier layer under the gate structure has a lower thickness than a second region of the barrier layer under the drain contact structure.
19. The semiconductor device of claim 18, wherein the first drain contact and the second metal portion of the second drain contact are parts of a contiguous metal contact that extends along the width direction of the semiconductor device.

- 20.** The semiconductor device of claim 18, the first drain contact extends along the width direction of the semiconductor device; and the second metal portion of the second drain contact extends along the width direction of the semiconductor device and is between the first drain contact and the first metal portion of the second drain contact.
- 21.** The semiconductor device of claim 18, wherein: the first drain contact extends along the width direction of the semiconductor device; a first region of the second metal portion of the second drain contact extends along the width direction of the semiconductor device and is between the first drain contact and the first metal portion of the second drain contact; and one or more second regions of the second metal portion of the second drain contact are separated by the first metal portion of the second drain contact.
- 22.** The semiconductor device of claim 18, wherein: the second metal portion of the second drain contact extends along the width direction of the semiconductor device; and the first drain contact includes a plurality of regions interspersed in and surrounded by the second metal portion of the second drain contact.
- 23.** The semiconductor device of claim 1, wherein the semiconductor layer is a first semiconductor layer, and the gate structure comprises: a second semiconductor layer over the barrier layer; and a gate contact layer over the second semiconductor layer.
- 24.** The semiconductor device of claim 23, wherein: the second semiconductor layer includes a third semiconductor portion and a fourth semiconductor portion; and the gate contact layer includes a third metal portion and a fourth metal portion, the third metal portion and the third semiconductor portion forming a third junction having a third energy barrier height, and the fourth metal portion and the fourth semiconductor portion forming a fourth junction having a fourth energy barrier height lower than the third energy barrier height.
- 25.** The semiconductor device of claim 23, wherein the second semiconductor layer is thicker than the first semiconductor layer.
- 26.** The semiconductor device of claim 1, wherein the gate structure comprises: a dielectric layer over the barrier layer; and a conductive layer over the dielectric layer.
- 27.** A method comprising: forming a patterned semiconductor layer over a barrier layer, the barrier layer being over a channel layer formed on a semiconductor substrate, the patterned semiconductor layer including a plurality of regions, each of the plurality of regions including a first semiconductor portion and a second semiconductor portion; forming first metal contacts on regions of the channel layer and the first semiconductor portions of the plurality of regions of the patterned semiconductor layer, in which the first semiconductor portions and the first metal contacts on the first semiconductor portions form first junctions; and depositing second metal contacts on the second semiconductor portions of the plurality of regions of the patterned semiconductor layer, in which the second semiconductor portions and the second metal contacts on the second semiconductor portions form second junctions having greater energy barrier heights than the first junctions.
- 28.** The method of claim 27, wherein: the second junctions include first Schottky junctions; and the first junctions include ohmic junctions, second Schottky junctions having energy barrier heights lower than the energy barrier heights of the second junctions, or a combination thereof.
- 29.** The method of claim 27, further comprising, before forming the first metal contacts: depositing a passivation layer on the barrier layer and the patterned semiconductor layer; and selectively etching the passivation layer and the barrier layer to expose the regions of the channel layer and the first semiconductor portions of the plurality of regions of the patterned semiconductor layer.
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